Creative Lighting Effect Design

Quickly Get Started With ARGB Lighting MCU Development

Henry

2024.09.10



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Agenda

- Nuvoton Lighting Ecosystem
- Evaluation Kit
- LED Lighting Strip Interface (LLSI)
- Sample Code

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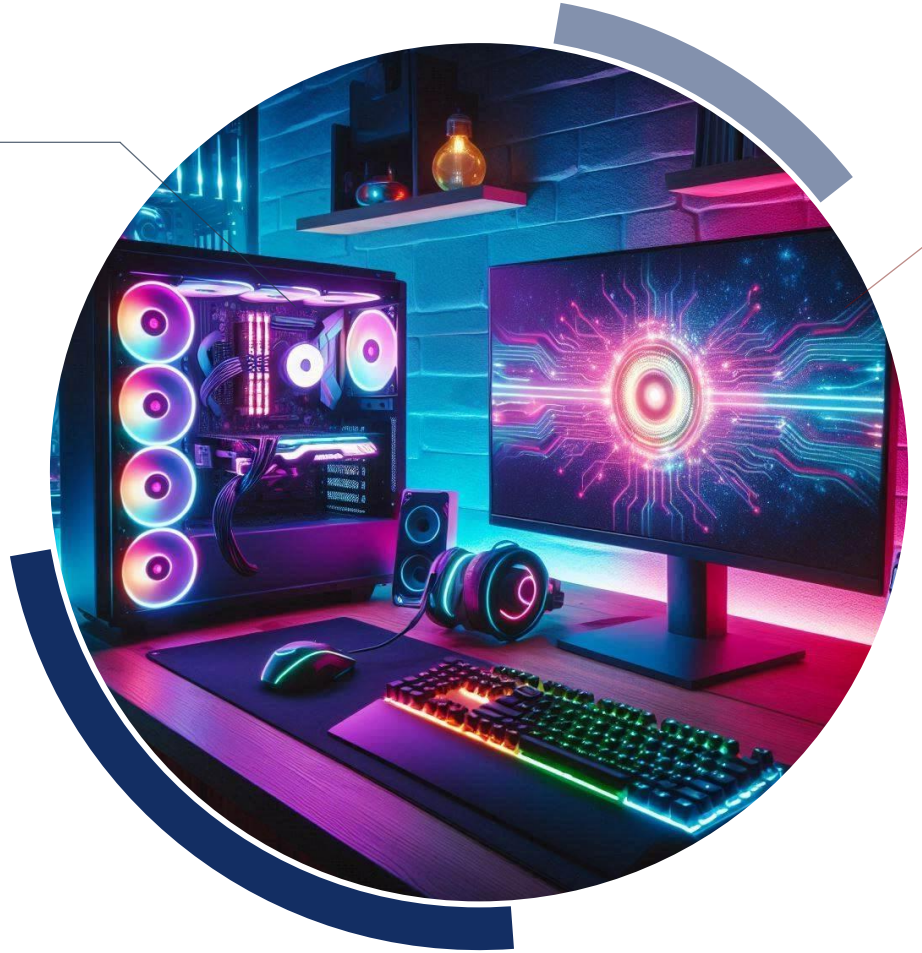
Nuvoton
Lighting Ecosystem



ARGB Lighting Application

PC Gaming

- CASE
- Mother Board
- CPU Cooler
- FAN
- Power Supply Unit
- SSD
- Memory(DDR4, DDR5)



Peripherals and Other Accessories

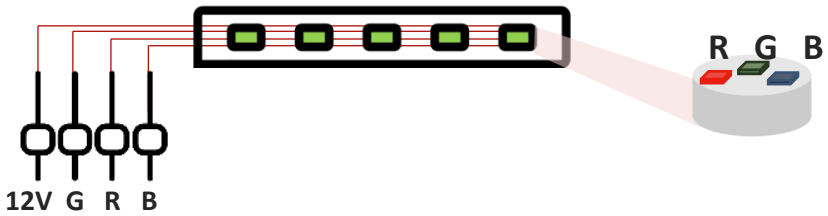
- Monitor
- Keyboard
- Mouse
- Mouse pad
- Headphone
- ARGB Cable
- LED Strip
- LED Stand

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The picture is generated by Copilot

LED Strip

RGB Strip



The entire strip is controlled together, with the color and brightness of the LEDs adjusted using **PWM**.

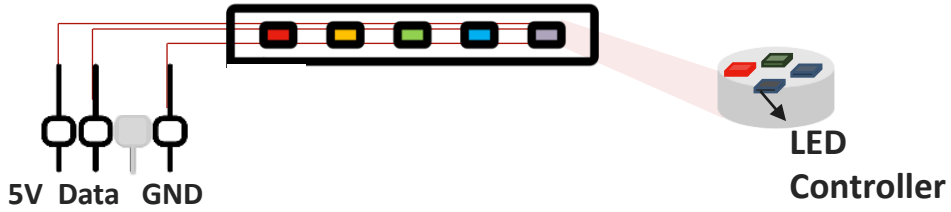


Connector: 12V, 4-pin



Single lighting effects

ARGB Strip
(Addressable RGB)



Each LED is controlled individually via a **data signal**



Connector: 5V, 3-pin



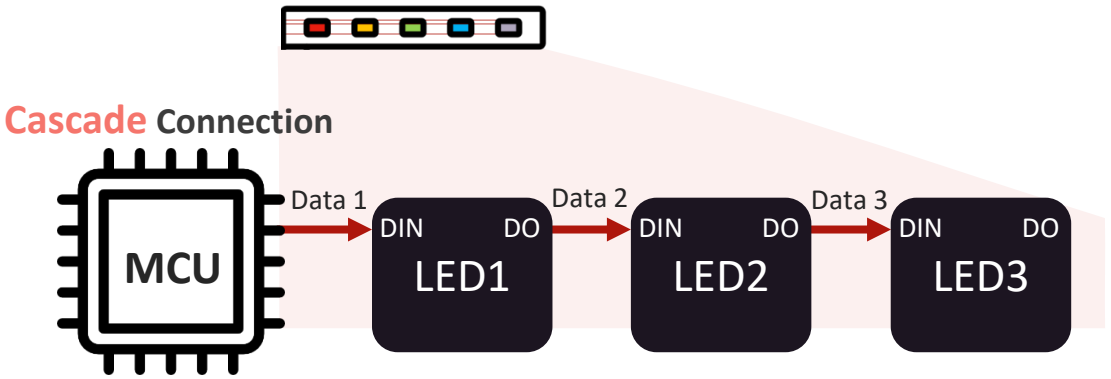
Advanced lighting effects

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Pictures are from Amazon.

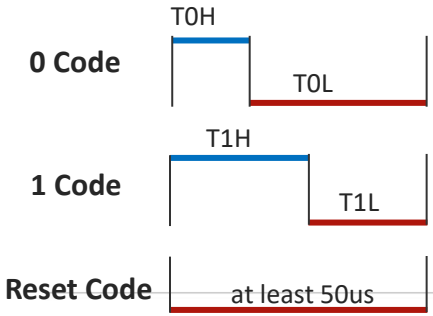
Control Method of ARGB

WS2812 Data Transmission Method



The data for each LED is sent sequentially. Each 24-bit frame is sent with the timing as described for each bit ('0' or '1').

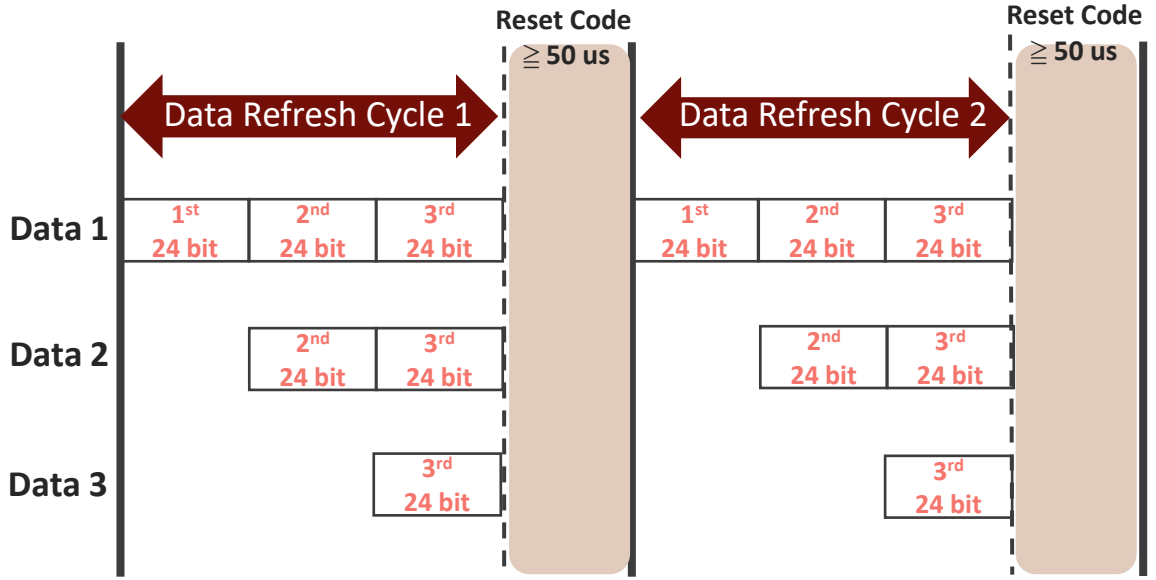
Sequence Timing



Data Frame Structure

24 bits Data Frame		
[G0:G7]	[R0:R7]	[B0:B7]

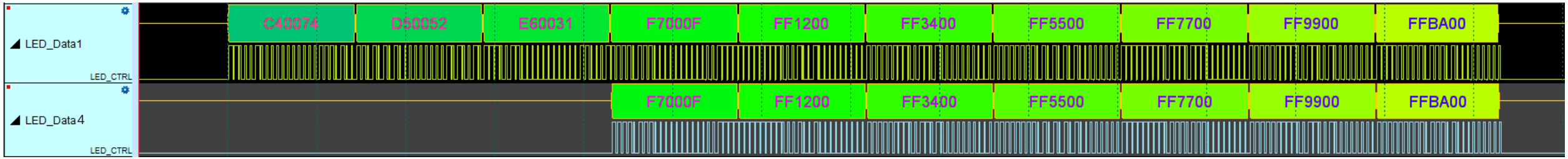
The 1st LED will receive the first 24 bits of color data, the 2nd LED will receive the second 24 bits of color data, and so on. Send a low-level signal for at least 50us to ensure the LED strip enters an idle state, ready to receive the next set of data.



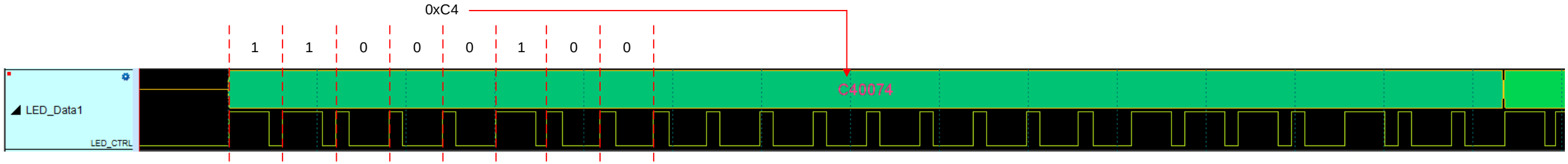
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ARGB LED

- Data Waveform
  - 10 pcs ARGB LED



  - Single ARGB LED data



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MCU and ARGB Strips

How does an MCU control ARGB strips?

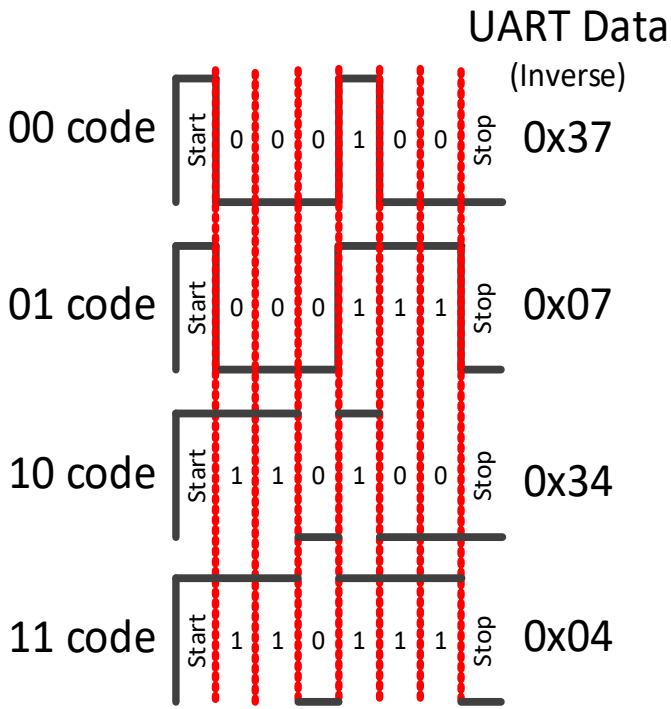
Interface	UART	SPI	PSIO (Programmable Serial I/O)	LLSI (LED Light Strip Interface)
			Nuvoton Patented	Nuvoton Exclusive's Hardware
RAM Usage (For 1 LED)	12 Bytes	12 Bytes	9 Bytes	3 Bytes
Data Mapping	Software	Software	Software	Hardware
Features & Limitations	Common interface • Depends on baud rate calculated • Need to consider start and stop bit data mapping • Inverse UART TX	Common interface Easier than UART • Depends on SPI rate calculated • Need to consider data format mapping	User-defined interface • Need to define the waveform by software	Hardware decode directly • Support period configuration for different strip's spec • Support RGB or GRB format
Nuvoton Supported	NuMicro® Family	NuMicro® Family	NuMicro® M251/252 Series	NuMicro® NUC1262/1263 Series

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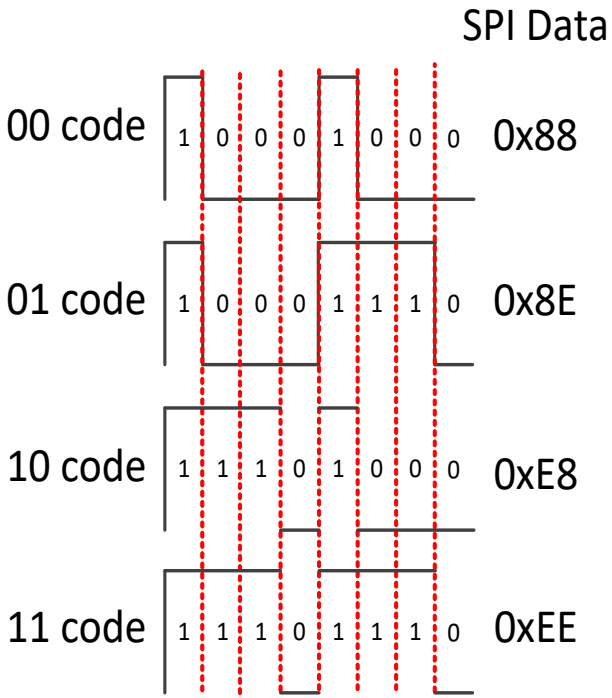
MCU and ARGB Strips

Data Mapping Example

UART

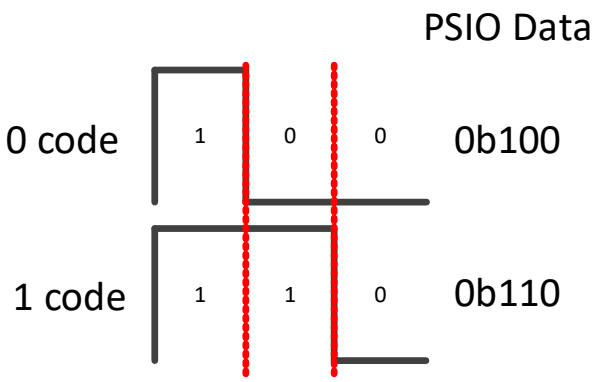


SPI



PSIO

(Nuvoton Patented)

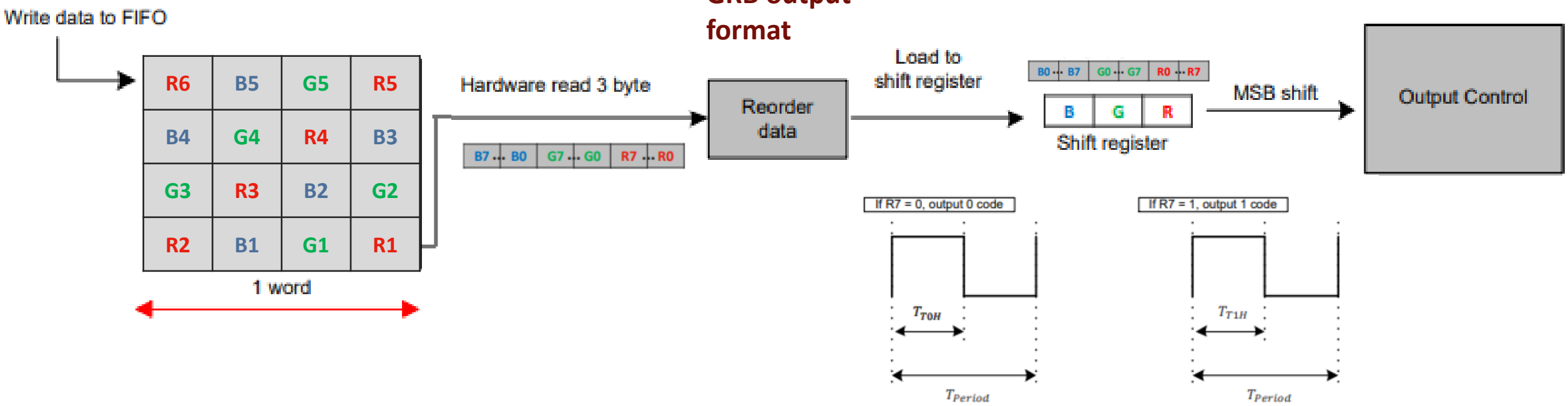


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MCU and ARGB Strips

Data Mapping Example

LLSI
(LED Light Strip Interface)



✓ Support RGB or GRB output format

✓ Support configurable T0H and T1H duty cycle

T_{T0H} = LLSI_CLK * (DIVIDER+1) * T0H
T_{T1H} = LLSI_CLK * (DIVIDER+1) * T1H
T_{Period} = LLSI_CLK * (DIVIDER+1) * PERIOD

Nuvoton Exclusive's Hardware

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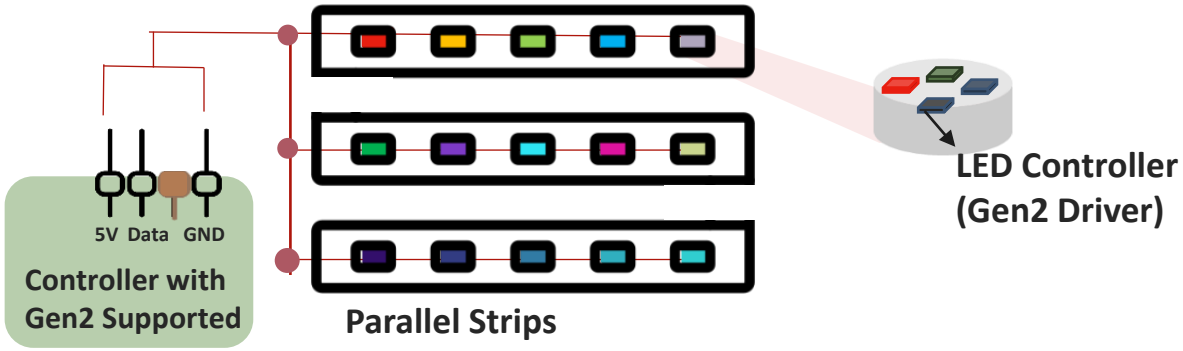
ARGB Gen1 and ARGB Gen2

Features

Gen2

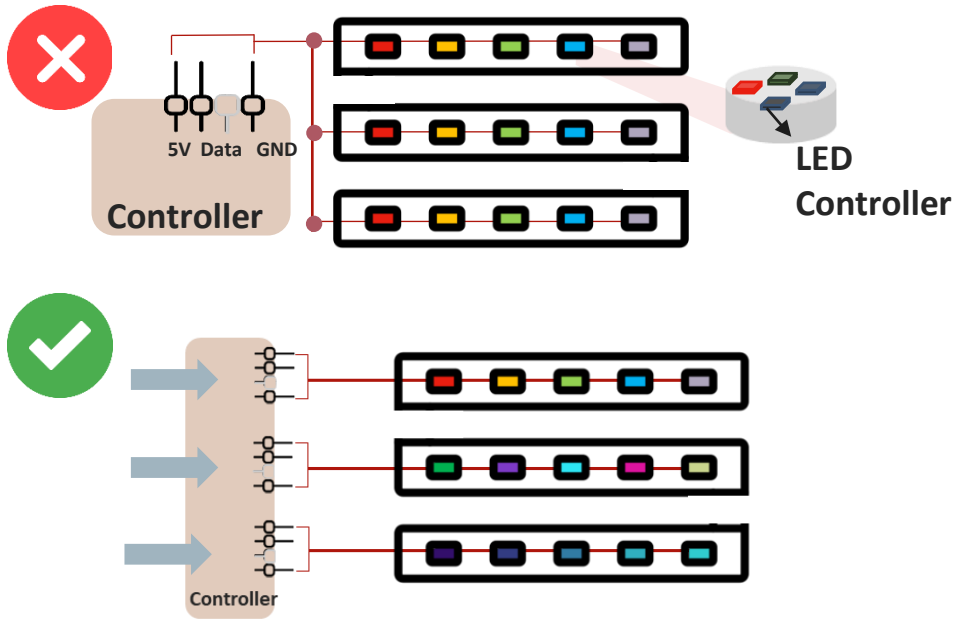
1 Port Control Parallel Strips

Different color across the parallel strips



Gen1

Need more ports to control each strips for different light effects.



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Nuvoton Lighting Ecosystem

NuMicro® Microcontroller Lighting Platform

NuMicro® NUC1262/NUC1263 Series

Algorithm of Lighting Effect

- Support 12+ lighting effects

Reference Design



Microsoft Windows11 Dynamic Lighting

- Support USB device and VHF driver
- Firmware integration

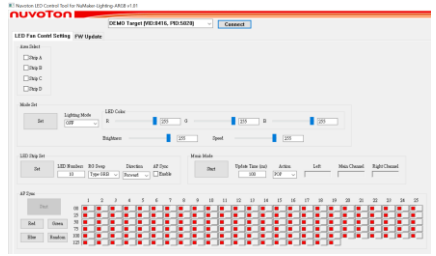


Mainstream Lighting Software Tools

- Certified FW and MCU for DDR5 RAM
- Supported list
  - ✓ Asus
  - ✓ GIGABYTE
  - ✓ MSI
  - ✓ ASROCK

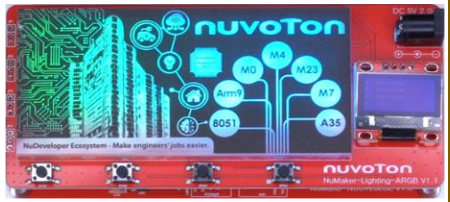
Lighting Control Testing Tool

- Open-source code
- Functions
  - ✓ Lighting mode setting
  - ✓ ISP FW upgrade



Evaluation Kit

- Board Support Package(BSP)
- Driving sample code- SPI, UART, LLSI, PSIO
- Gen1/Gen2 sample code



NuMaker-Lighting-ARGB

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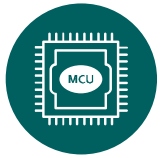
NuMicro® NUC1262/NUC1263 Series MCU

Flash Memory/ SRAM (Kbytes)	QFN33 (5 x 5 mm) Pitch 0.5 mm	QFN48 (7 x 7 mm) Pitch 0.5 mm	LQFP48 (7 x 7 mm) Pitch 0.5 mm	LQFP64 (7 x 7 mm) Pitch 0.4 mm
128 / 20		NUC1262NE4AE	NUC1262LE4AE	NUC1262SE4AE
64 / 20	NUC1263ZD4AE	NUC1263ND4AE	NUC1263LD4AE	NUC1263SD4AE

- **Arm® Cortex®-M23 core**
- **Runs up to 72 MHz**
- **Operating voltage:**
  2.5V ~ 5.5 V
- **Operating temperature:**
  -40°C ~ +105°C
- **High immunity:**
  EFT 4.4 kV, ESD HBM 7 kV
- LLSI (LED Light Strip Interface)
  NUC1262: 10-channel
  NUC1263: 6-channel
- NUC1263 supports 2 sets of I^3C, 4 sets of ACMP, 4 sets of DAC

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NuMicro® NUC1262 Series Features



System

- Arm® Cortex®-M23 core
- Runs up to 72 MHz
- 128 KB Flash Memory, 20 KB SRAM
- Operating voltage: 2.5V ~ 5.5V
- Operating temperature: -40°C ~ +105°C
- 4 KB Loader ROM (LDROM)
- 10-channel PDMA



Peripherals

- **Up to 10-channel LLSI (LED Light Strip Interface)**
- Crystal-less USB 2.0 FS Device
- 2 sets of UART
- 2 sets of I^2C
- 2 sets of SPI/I^2S
- 24-channel 16-bit 72 MHz PWM
- 9-channel 50 mA High sink I/O



Analog

- 8-channel 12-bit 800 Ksps ADC



Security

- Up to 2048 bytes Security Protection ROM (SPROM)

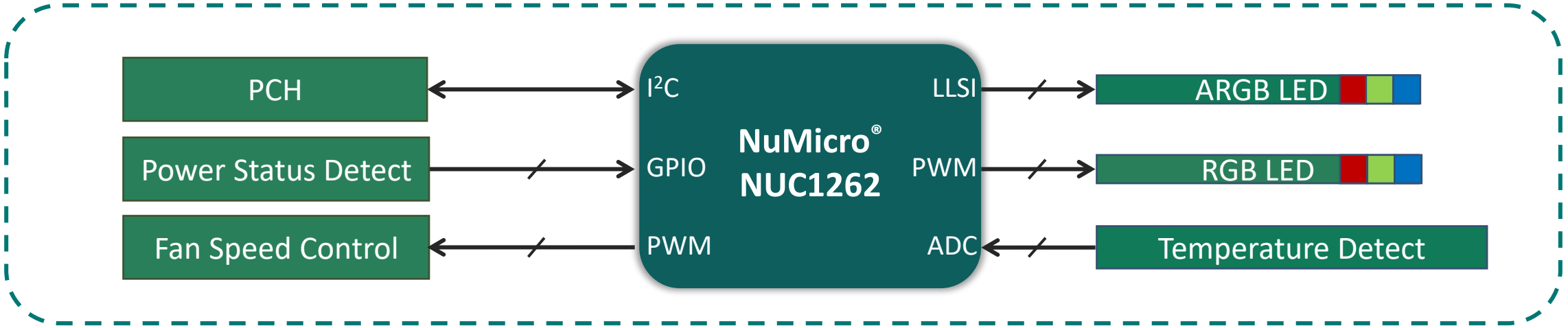
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Mother Board Gaming Lighting

Features

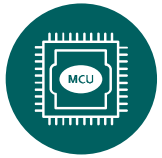
- LLSI controls ARGB LEDs
- PWM controls RGB LEDs
- Supports FAN control, temperature detect, etc.

Mother Board



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NuMicro® NUC1263 Series Features



System

- Arm® Cortex®-M23 core
- Runs up to 72 MHz
- 64 KB Flash Memory, 20 KB SRAM
- Operating voltage: 2.5V ~ 5.5V
- Operating temperature: -40°C ~ +105°C
- 4 KB Loader ROM (LDROM)
- 10-channel PDMA



Peripherals

- 6-channel LLSI
- Crystal-less USB 2.0 FS Device
- 3 sets of UART
- 3 sets of I^2C
- 2 sets of I^3C slave
- 3 sets of SPI/I^2S
- 24-channel 16-bit 144 MHz PWM



Analog

- Up to 16-channel 12-bit 800 Ksps ADC
- 4 sets of ACMP
- 4 sets of DAC



Security

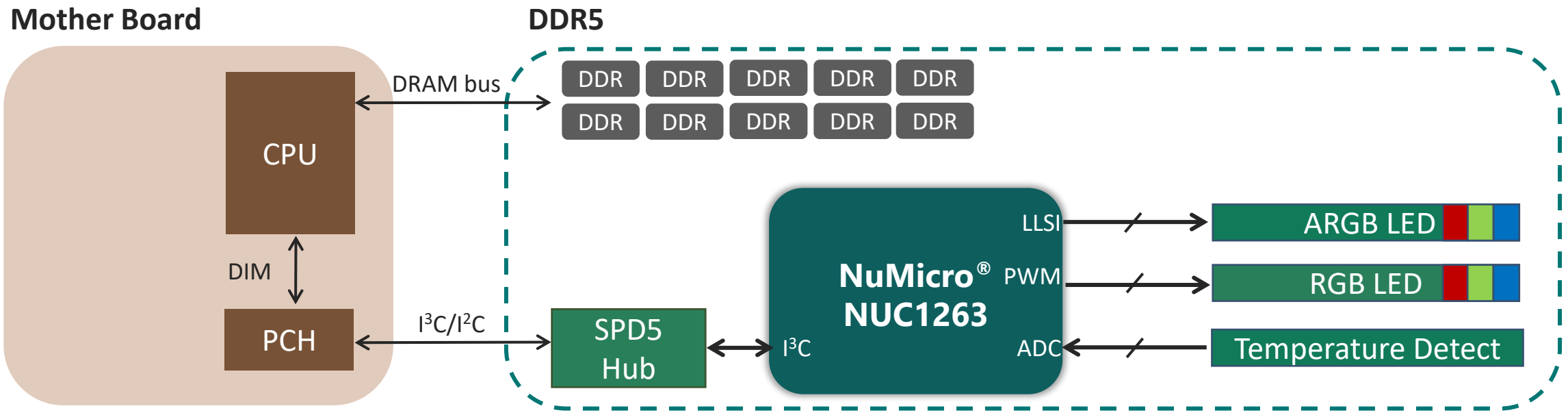
- Up to 2048 bytes Security Protection ROM (SPROM)

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DDR5 Gaming Lighting Control

Features

- Supports I³C slave interface
- LLSI control ARGB LEDs
- PWM control RGB LEDs
- **Patent** to sync lighting effect between different DDR5 modules



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FAQ

- How many ARGB LED can MCU control?
  - To ARGB LED, cascade number are not less than **1024 pcs** when the refresh rate is 30 fps. However, user must consider hardware limitation.
    - **Power consumption**: If each LED draws up to 60 mA at peak brightness, it takes 3 A when driving 50 pcs ARGB LED.
    - **Voltage drop**: The 5V power supply drops as the series ARGB LEDs are connected. You may need another 5V power supply on end side or multiple 5V power supply in the middle.
  - To LLSI, the max ARGB LED data transfer count is 4095 (PCNT (LLSI_PCNT[11:0])). It takes about 12 KB SRAM.
  - To another interfaces, it should consider the **SRAM usage**.

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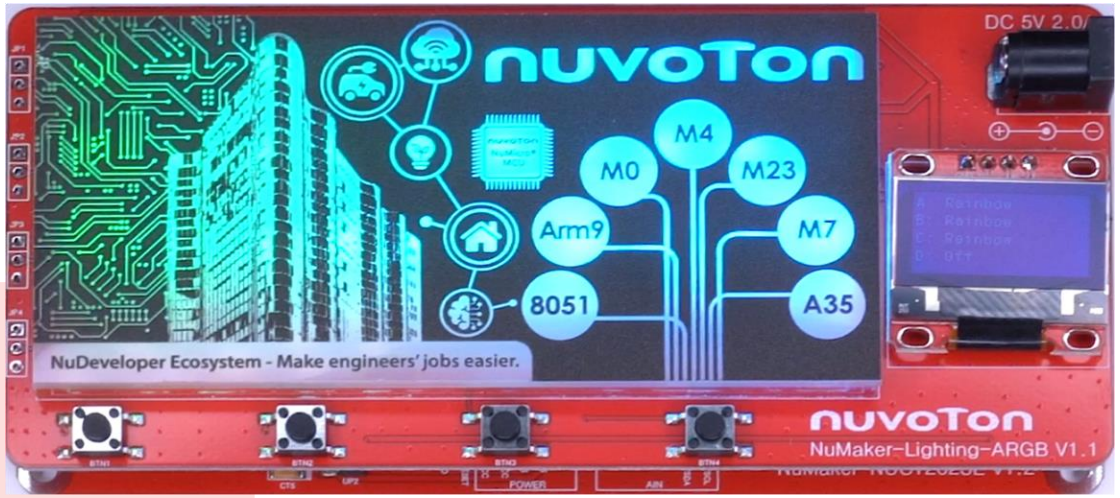
Evaluation Kit

NuMaker-Lighting-ARGB



ARGB-Lighting Control

Features



Order number: NK-ARGBLED

NuMaker-Lighting-ARGB
Evaluation Kit

Easy Solution for Advanced ARGB LED Lighting Control

Multi-Platform w/ Multi-Interface

- ✓ Compatible with multiple NuMaker Platforms
- ✓ Compatible Interfaces:

LLSI, PSIO, SPI, USCI-SPI, UART

(PSIO only for M252; UART only for Gen1 control)

Flexible Development

- ✓ On-board **40 pcs Gen2 ARGB LED** (4 strips, 10 pcs each)
- ✓ Supports both **Gen1 and Gen2** sample code
- ✓ Supports **ARGB LED extension header** and 5V external power jack to extend more LED strips

Software Tool Support

- ✓ Support **Microsoft Dynamic Lighting** control (Windows 11)
- ✓ Support **PC tool** to control lighting effects via **USB interface**

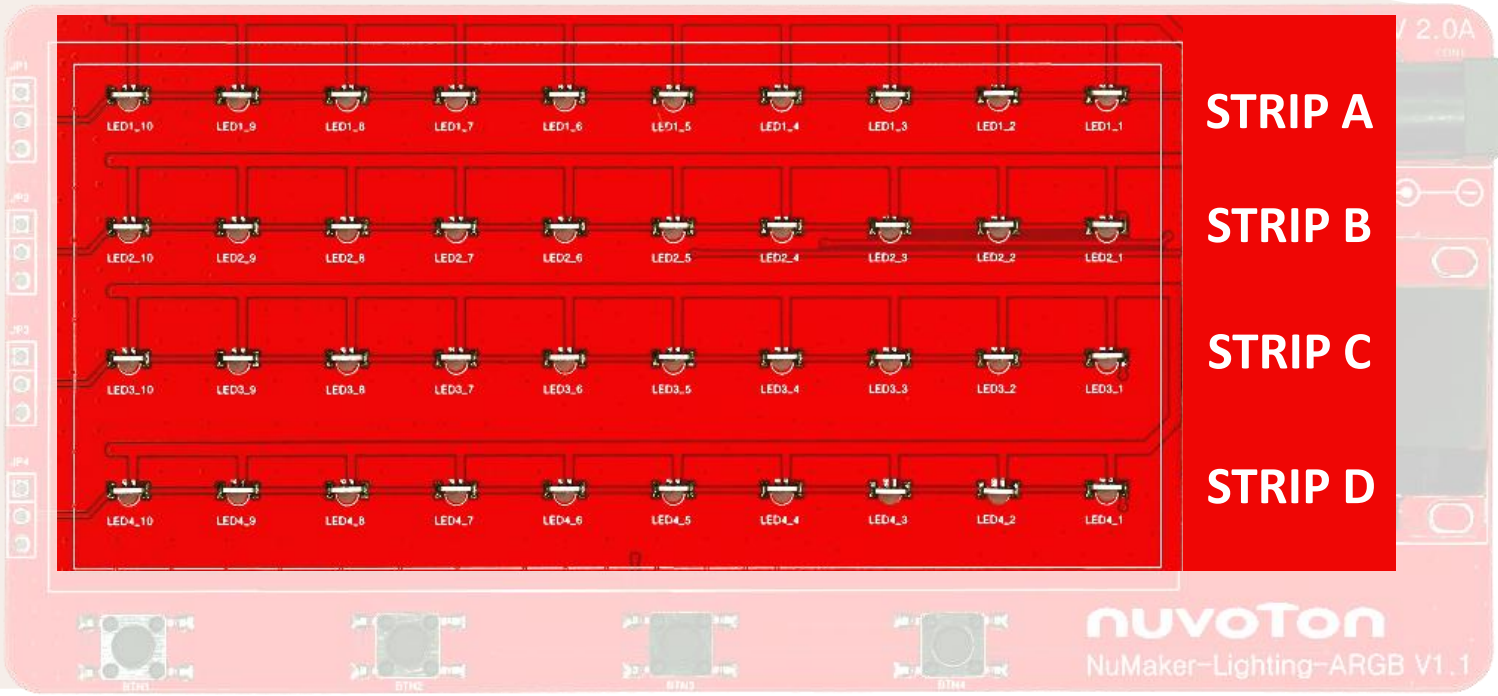
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NuMaker-Lighting-ARGB Development Kit

Components & Functions



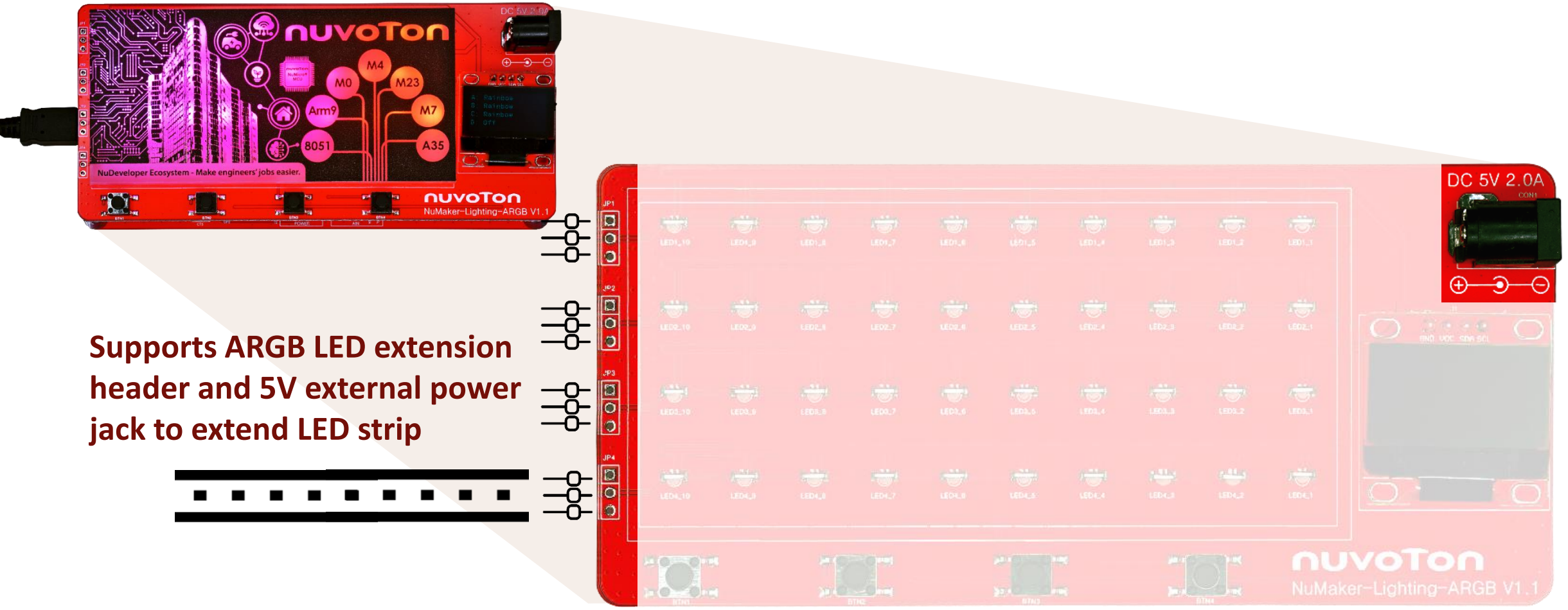
Supports totally 4 x 10 pcs Gen2 ARGB LED



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NuMaker-Lighting-ARGB Development Kit

Components & Functions



Supports ARGB LED extension header and 5V external power jack to extend LED strip

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NuMaker-Lighting-ARGB Development Kit

Components & Functions



An OLED panel(0.96”, resolution 128x64) to show current lighting mode of each strip



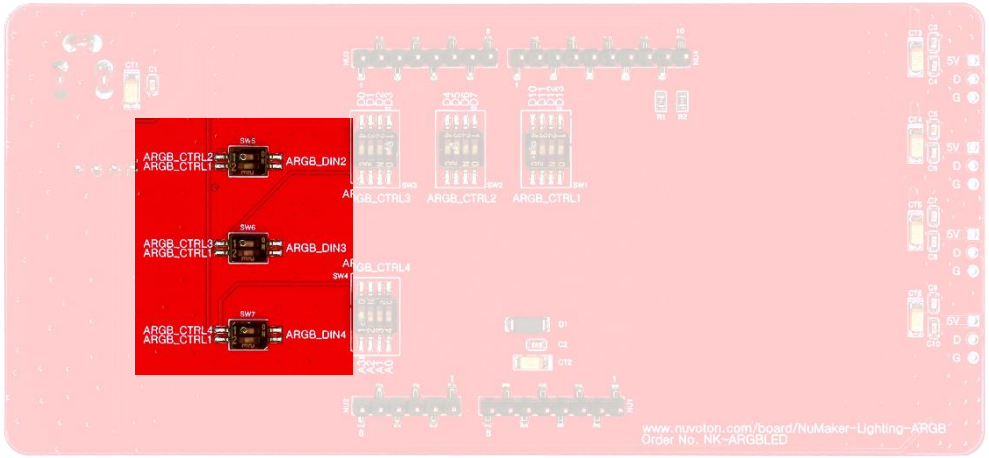
4 Buttons to change lighting mode for each strip

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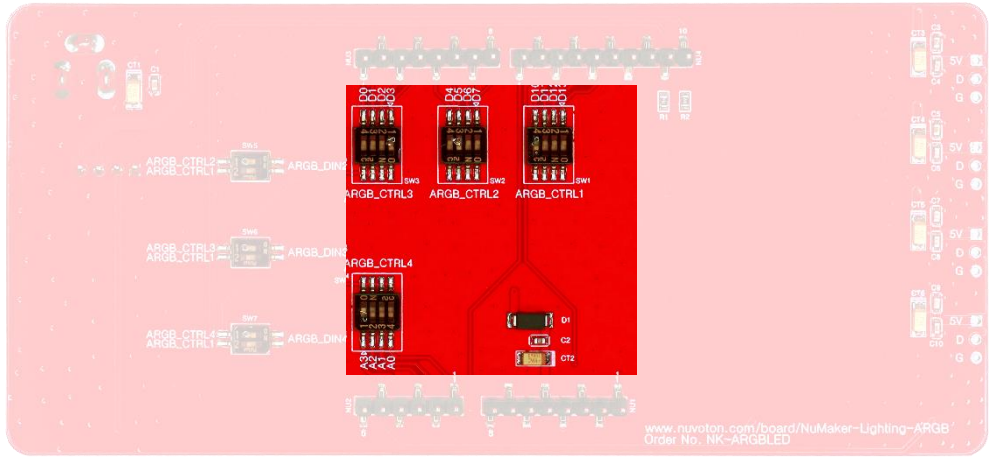
NuMaker-Lighting-ARGB Development Kit

Components & Functions

Gen1 or Gen2 Function Switch



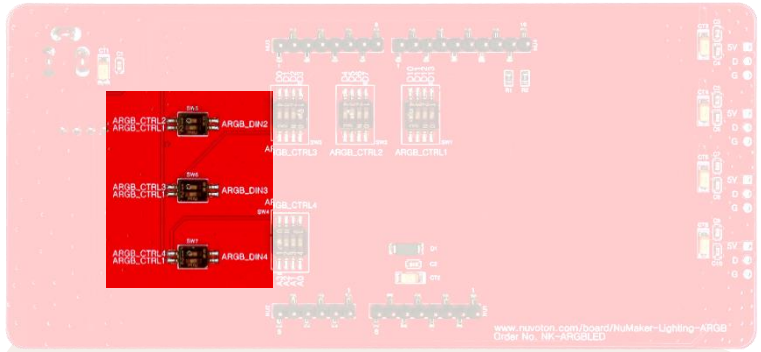
Pin Definition Switch for Different Board



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Switch Status

Gen1 or Gen2 Function



Back View – NuMaker-Lighting-ARGB

Gen1

SW5
CTRL 2
CTRL 1

SW6
CTRL 3
CTRL 1

SW7
CTRL 4
CTRL 1

ON —
OFF —

Gen2

SW5
CTRL 2
CTRL 1

SW6
CTRL 3
CTRL 1

SW7
CTRL 4
CTRL 1

ON —
OFF —

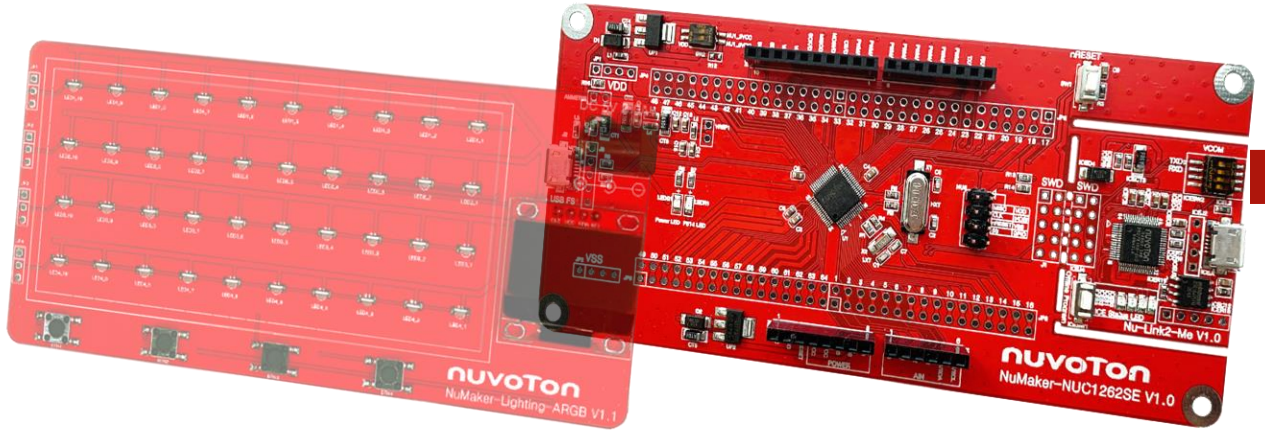
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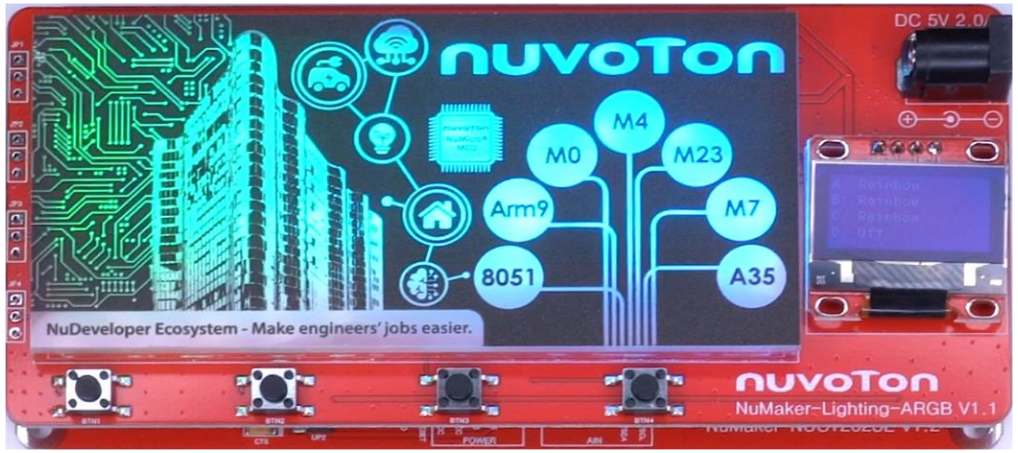
List of Compatible NuMaker Platforms

NuMaker-Lighting-ARGB Evaluation Kit

NuMaker Platforms

- NuMaker-NUC1262SE
- NuMaker-NUC1263SD
- NuMaker-M252SD
- NuMaker-M032SE
- NuMaker-M467

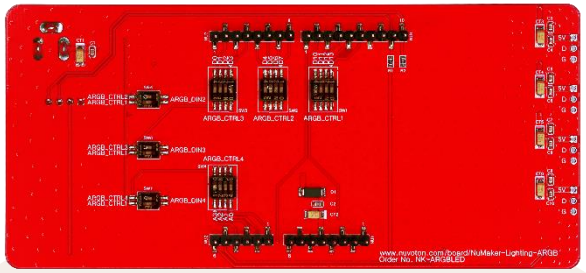




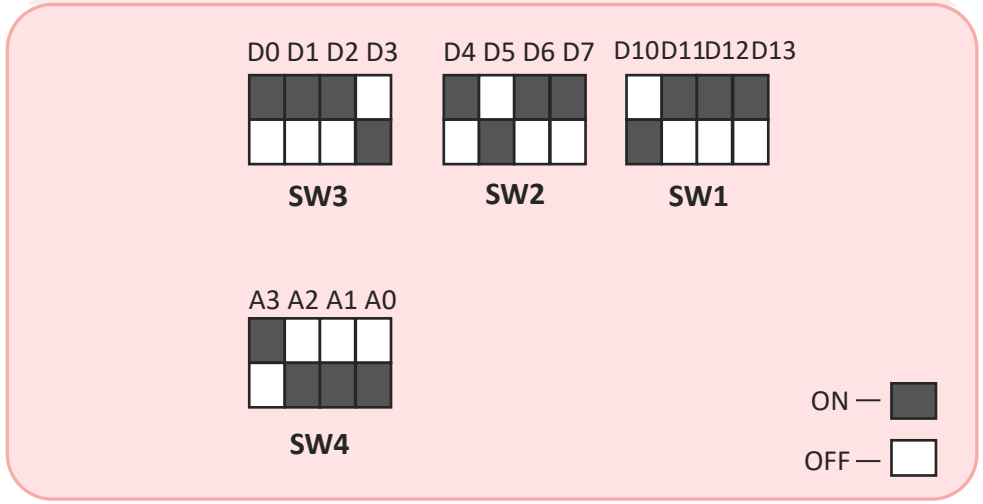
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Switch Status

NuMaker-NUC1262SE + NuMaker-Lighting-ARGB



Back View – NuMaker-Lighting-ARGB

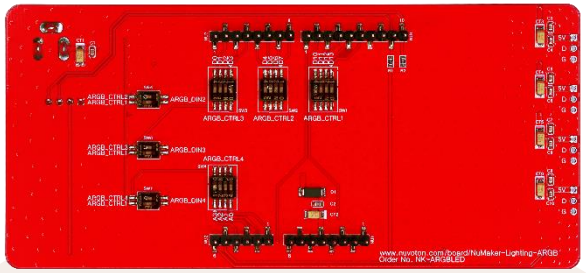


STRIP	Interface	NUC1262SE Pin Number	Arduino Uno Pin Name
1	LLSI4	PA3	D10
2	LLSI3	PC2	D5
3	LLSI0	PC5	D3
4	LLSI5	PB4	A3

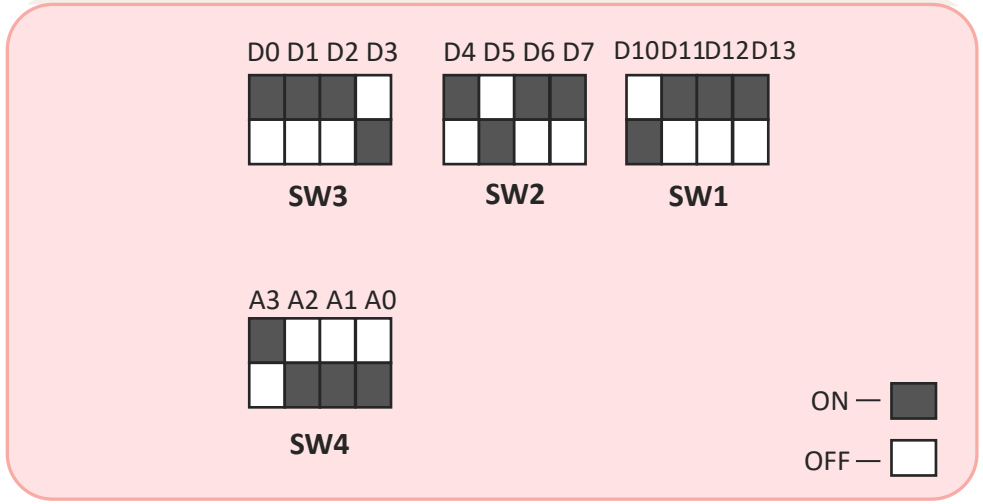
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Switch Status

NuMaker-NUC1263SD + NuMaker-Lighting-ARGB



Back View – NuMaker-Lighting-ARGB

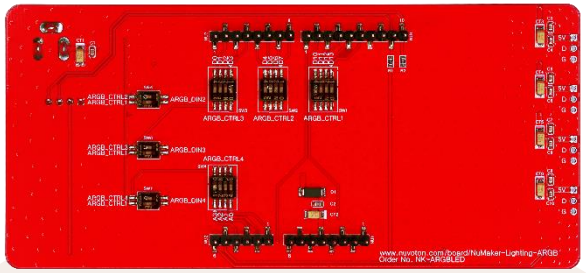


STRIP	Interface	NUC1263SD Pin Number	Arduino Uno Pin Name
1	LLSI4	PA3	D10
2	LLSI3	PC2	D5
3	LLSI0	PC5	D3
4	LLSI5	PB4	A3

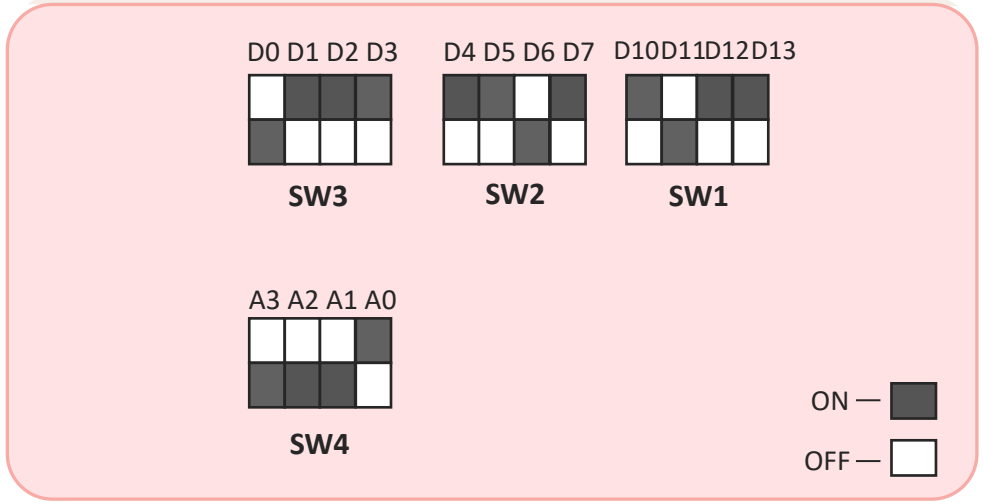
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Switch Status

NuMaker-NUC1261SG + NuMaker-Lighting-ARGB



Back View – NuMaker-Lighting-ARGB

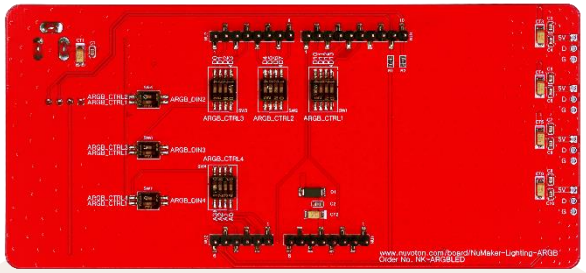


STRIP	Interface	NUC1261SG Pin Number	Arduino Uno Pin Name
1	SPI0 MOSI	PE11	D11
2	USCI0 DATA0	PC0	D6
3	USCI1 DATA0	PB0	D0
4	SPI1 MOSI	PB5	A0

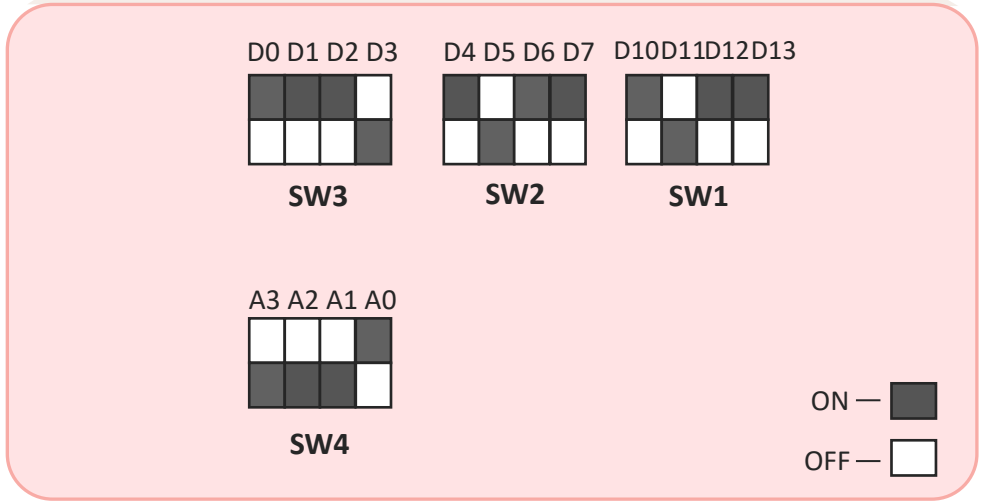
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Switch Status

NuMaker-M252SD + NuMaker-Lighting-ARGB



Back View – NuMaker-Lighting-ARGB

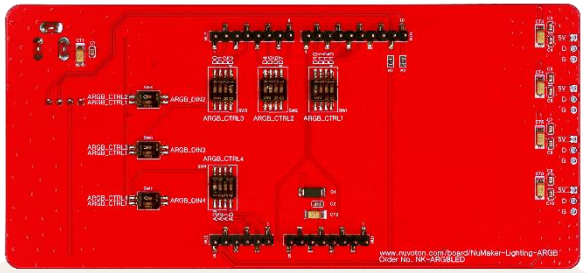


STRIP	Interface	M252SD Pin Number	Arduino Uno Pin Name
1	SPI0 MOSI	PA0	D11
2	PSIO0 CH3	PC2	D5
3	PSIO0 CH0	PC5	D3
4	USCI1 DATA0	PB7	A0

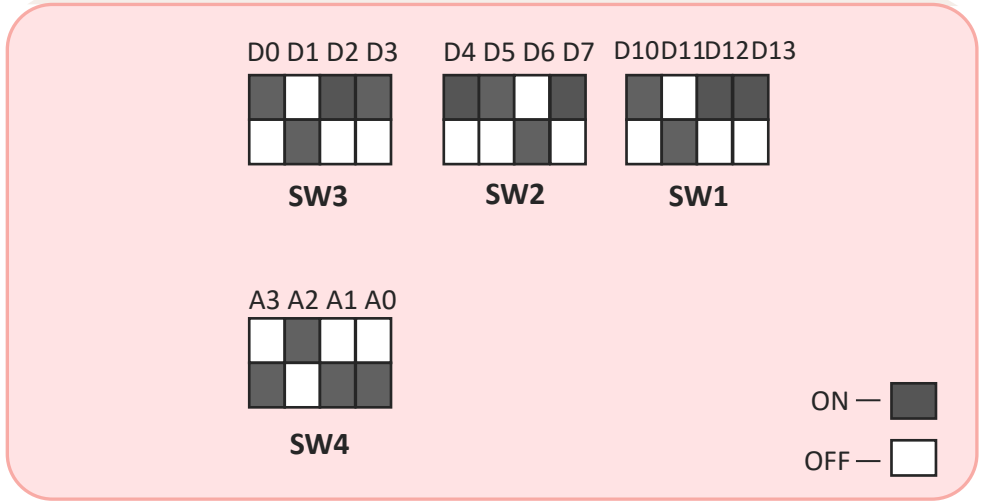
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Switch Status

NuMaker-M032SE + NuMaker-Lighting-ARGB



Back View – NuMaker-Lighting-ARGB

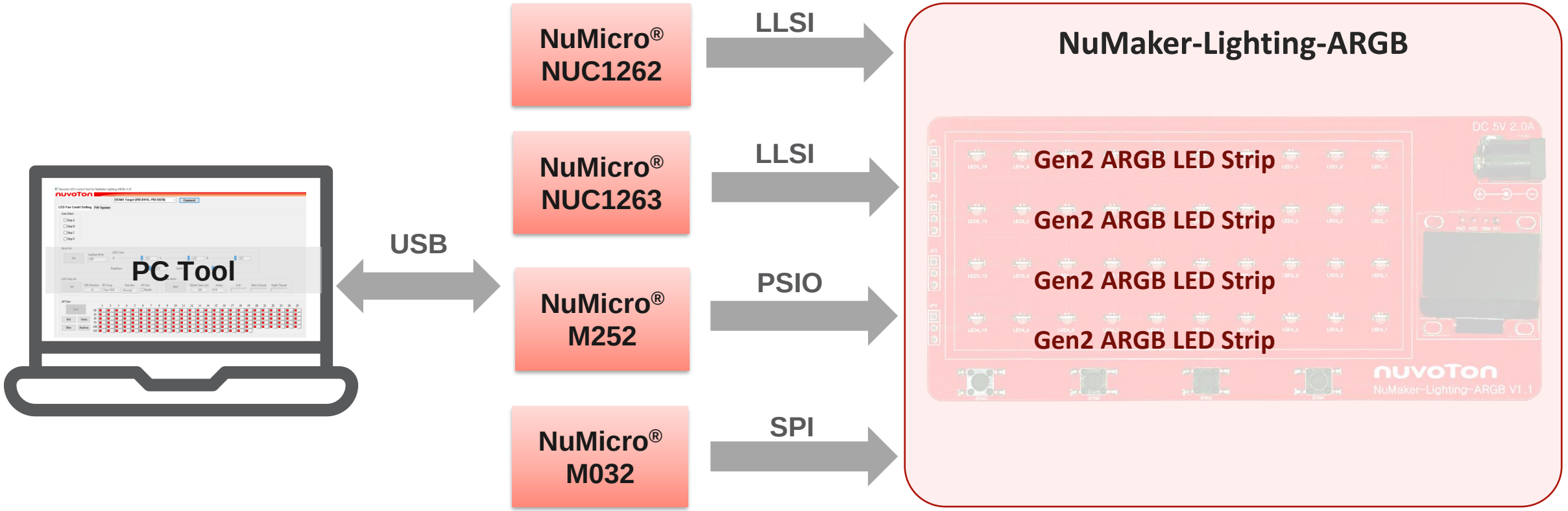


STRIP	Interface	M032SE Pin Number	Arduino Uno Pin Name
1	SPI0 MOSI	PA0	D11
2	UART0 TX	PA7	D6
3	UART1 TX	PB3	D1
4	UART2 TX	PB5	A2

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NuMaker-Lighting-ARGB Development Kit

Block Diagram

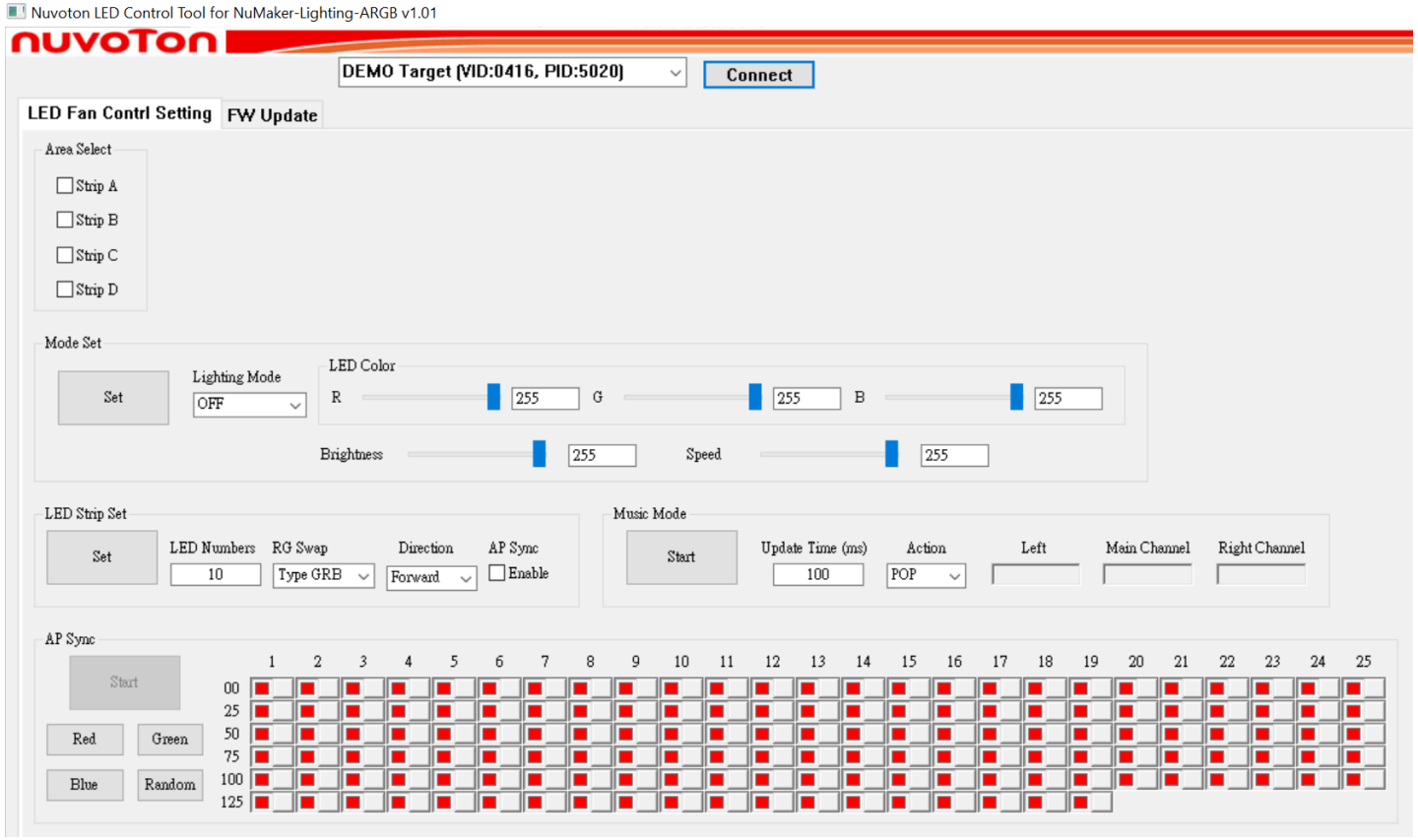


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Lighting Control Testing Tool

NuLEDCtrl

- Select Target Strip
- Set Lighting Setting
  - Mode / Color / Brightness / Speed
- Set Strip Setting
  - Number / Type / Direction / AP Sync
- Enable music mode
  - Update Time
  - Action Type
    - POP / Jazz / Mixed
- AP Sync Control
  - Direct control LED color by tool



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Lighting Effects

Algorithm Supported

LED Mode	Description
OFF	All LEDs are off.
STATIC	All LEDs show the static color.
BREATHING	All LEDs show the breathing effect.
STROBE	All LEDs show the on, off effect.
CYCLING	All LEDs show the color in sequence of Red, Green and Blue.
RANDOM	All LEDs show the color in random of Rainbow color. Red, Orange, Yellow, Green, Cyan, Blue, Indigo and Purple.
MUSIC_POP	All LEDs show the color from Blue to Red with gradient based on volume value.
MUSIC_JAZZ	Lighten LED number based on volume value.
MUSIC_MIXED	Lighten LED number based on volume value and so do the color from Blue to Red.
WAVE	All LEDs show the color in sequence of Rainbow color with gradient.
SPRING	Meteor effect in forward direction, and then do in backward direction.
WATER	Every 4 LEDs show the color in brightness ration of [100%, 15%, 5%, 15%]. And all LEDs show the color which changes in sequence of Rainbow color.
RAINBOW	Shows the color in sequence of Rainbow color with gradient and the color gradient also moves through LEDs.

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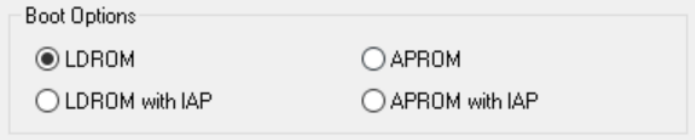
NuMaker-Lighting-ARGB Reference Resource

- Document
  - Quick Start
  - User manual
  - Schematic
- Resource
  - \Resource\Source Code: Sample code and NuLEDCtrl tool source code.
  - \Resource\Tool: NuLEDCtrl tool to control lighting effect through USB interface.
  - \Resource\Program File: ICP project and bin file for programming.

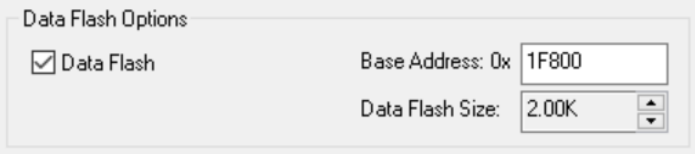
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Program NuMaker-Lighting-ARGB

- Use ICP Programming Tool
  - LDROM: [Target Chip]_ISP_HID.bin
  - APROM: [Target Chip]_[Sample Code]_vxx_xx_**Program**.bin
  - Config Bits Setting:
    - Set **Boot Options** to LDROM.



    - Enable **Data Flash** and set Data Flash Size to 2 KB.



  - Select LDROM, APROM and Config in Programming area.
  - Click “Start” to program target chip.

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Update NuMaker-Lighting-ARGB

- Use NuLEDCtrl Tool
  - Connect MCU side's USB to PC.
  - Open PC control tool and click "Connect".
  - Change to "FW Update" page.
  - Click "APROM" and select [Target Chip]_[Sample Code]_vxx_xx_**Update**.bin file.
  - Click "Start" and wait for update finish.

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Successful Story



Successful Story

Microsoft Windows11 Dynamic Lighting



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Successful Story

Gaming Products



Gaming Monitor RGB LED
NUC1262SE4AE
NUC1262NE4AE
NUC1263SD4AE

Gaming HMI/DISPLAY
N9H20 / N9H26 / N9H30
MA35D1

Case, VGA Card, Cooler
NUC1263ND4AE
NUC1262SE4AE

MB NUC1263SD4AE
FAN MS51BA9AE
DDR5 NUC1263ZD4AE

CPU Cooler HMI (OLED/LCD)
N9H20 / N9H26 / N9H30
M487

HeadSet
ML51 / MS51

Gaming Key Board
NUC1262SE4AE

MICE NUC1262NE4AE/M032SE3AE/M032SG6AE
8K MICE M467KJHAN/M484KIDAE
GAMEPAD M252 / M031BT

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Successful Story

Gaming Products List



	CASES	KEYBOARD	HEADSETS	MONITORS	AUDIO
Nuvoton Platform	M467 M484 NUC1262 MS51	M467 M484 NUC1262 M032	MS51 ML51	M467 M484 NUC1262	MS51 ML51

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Successful Story

Gaming Products List



	MICE	POWER	CPU COOLER	CUSTOM	FAN
Nuvoton Platform	M467 M484 NUC1262 M032	M467 M484 NUC1262	N9H20 N9H26 N9H30 MA35D1	M467 M484 NUC1262	MS51 ML51

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Successful Story

Gaming Products List



	MEMORY	STORAGE	PADS	LIGHTING	DOCKS
Nuvoton Platform	NUC1263	NUC1262 NUC1263	MS51 ML51 M032	MS51 ML51	M467 M484

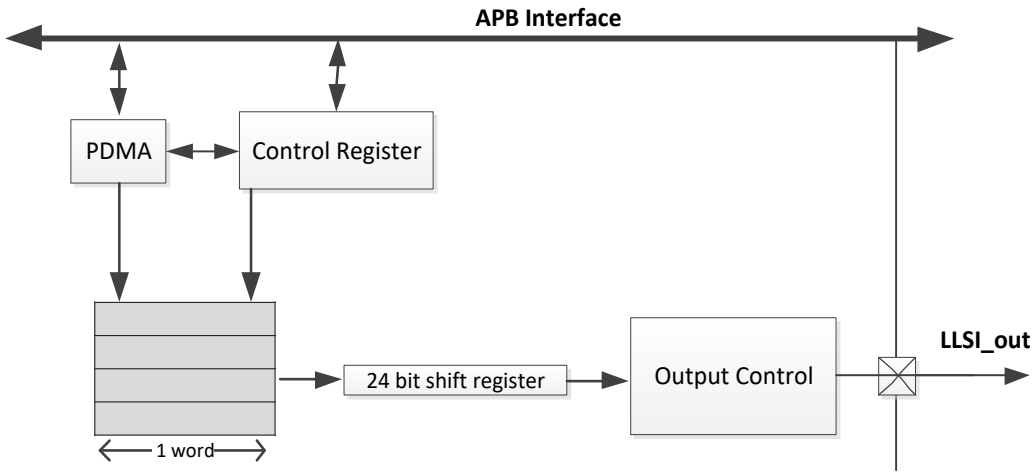
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LED Lighting Strip Interface (LLSI)



LED Lighting Strip Interface (LLSI)

- Features
  - 4x32-bit TX FIFO
  - Configurable transfer period and frame reset length
  - Configurable T0H and T1H duty cycle
  - Supports RGB and GRB output format
  - Supports Software mode and **PDMA** base mode transfer
- Block Diagram

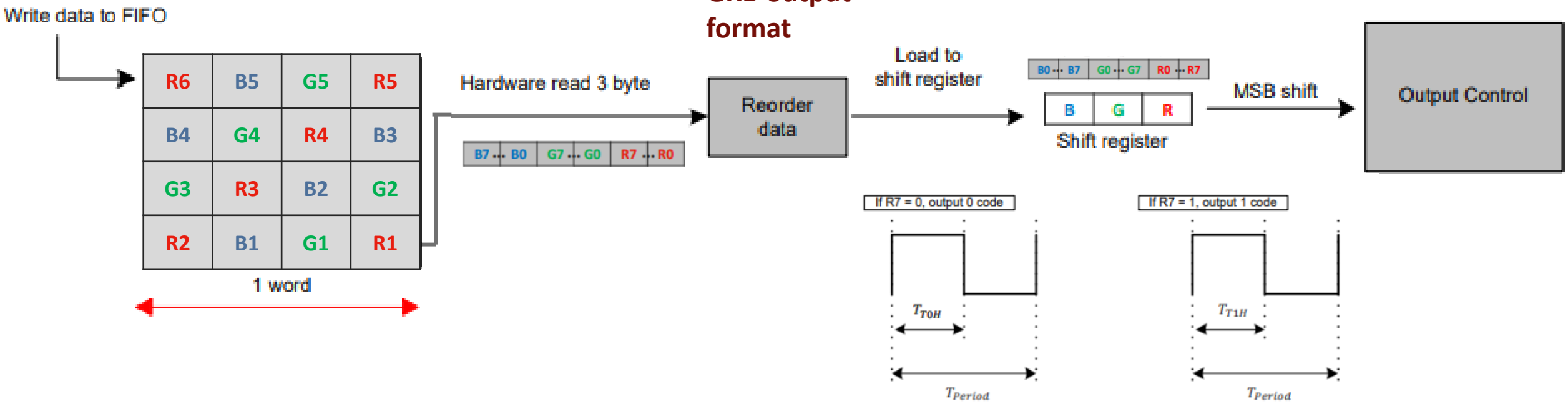


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LED Lighting Strip Interface (LLSI)

Data Mapping Example

LLSI
(LED Light Strip Interface)



✓ Support RGB or GRB output format

✓ Support configurable T0H and T1H duty cycle

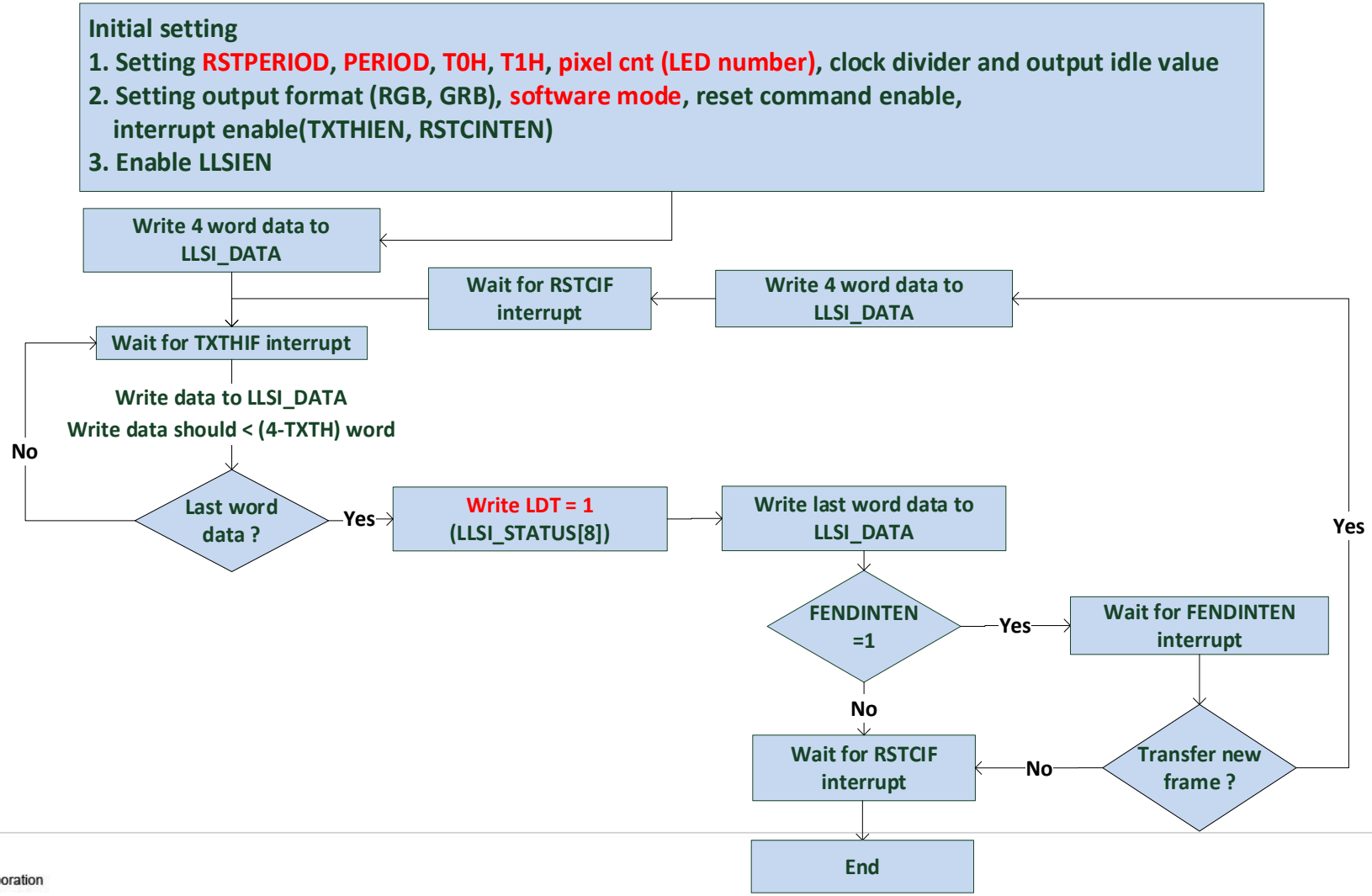
Nuvoton Exclusive's Hardware

T_{T0H} = LLSI_CLK * (DIVIDER+1) * T0H
T_{T1H} = LLSI_CLK * (DIVIDER+1) * T1H
T_{Period} = LLSI_CLK * (DIVIDER+1) * PERIOD

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LED Lighting Strip Interface (LLSI)

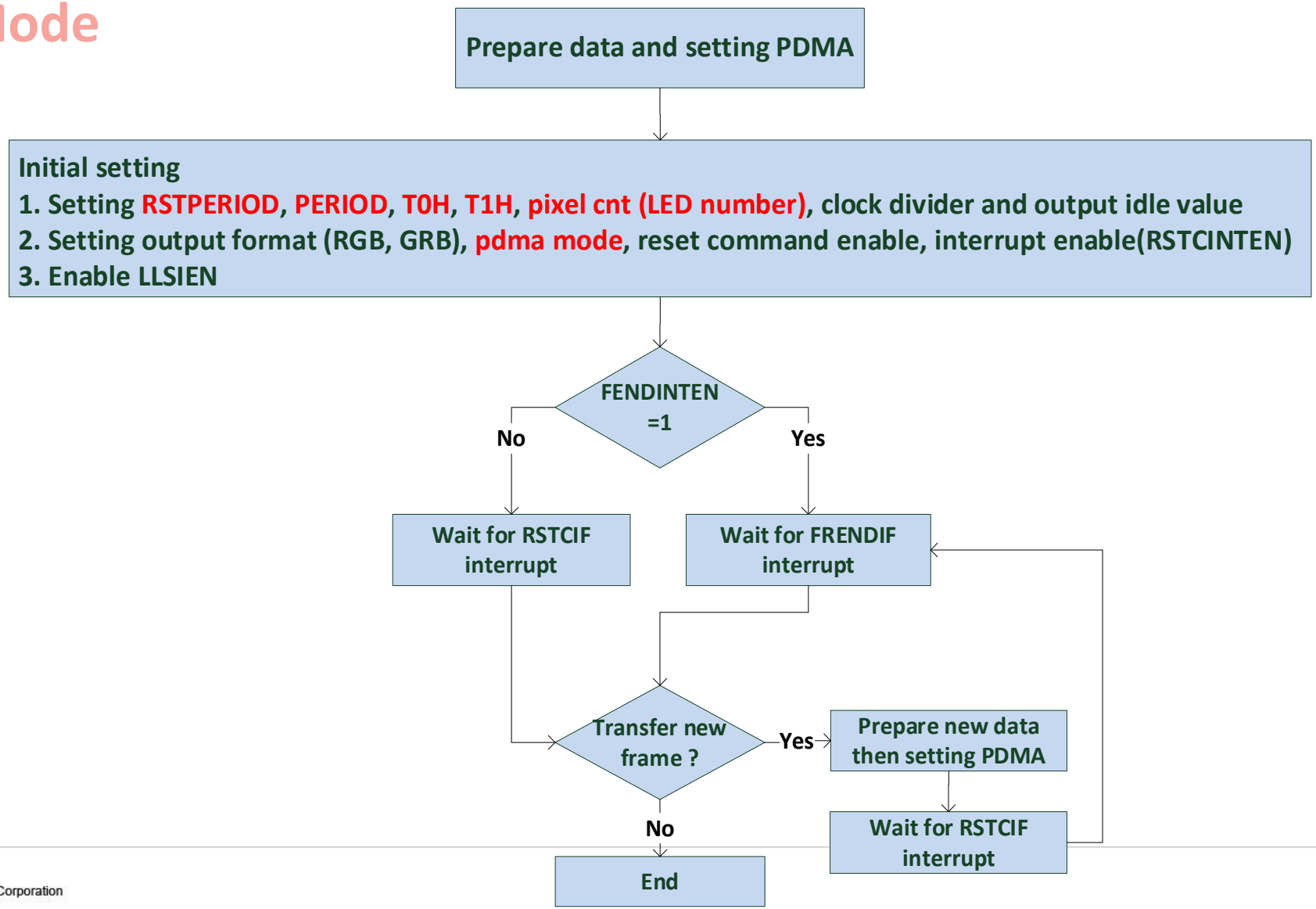
Software Mode



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LED Lighting Strip Interface (LLSI)

PDMA Mode



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LED Lighting Strip Interface (LLSI)

- API function (llsi.c & llsi.h)
  - LLSI_**Open**
    - Configure control setting of LLSI module, such as single bit time, T0H/T1H time, Reset Code time, LED count (pixel count).
  - LLSI_**SET_RGB_FORMAT**
    - Set data format as RGB or GRB.
  - LLSI_**ENABLE_RESET_COMMAND**
    - Enable hardware to send Reset Code after LED data.
  - LLSI_**EnableInt**
    - Enable LLSI interrupt, such as frame end interrupt, Reset Code interrupt.
  - LLSI_**SET_PDMA_MODE**
    - Set as PDMA mode to send request to PDMA module.
  - LLSI_**SET_SW_MODE**
    - Set as software mode, stop sending request to PDMA module.

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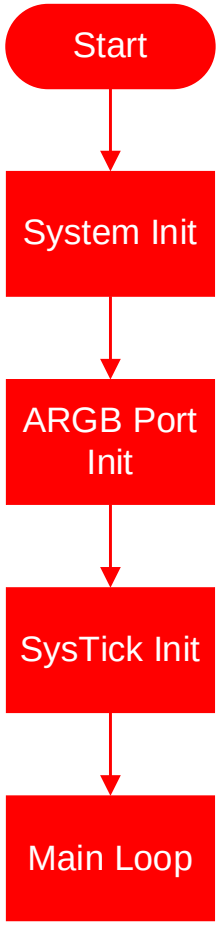
Sample Code

Use NuMaker-NUC1262SE



Sample Code

- Program Flowchart



- Initialize system clock
- Enable PDMA module
- Read LED setting from Data Flash

- Initialize LLSI and MFP (Multi-Function Pin)
- Initialize PDMA
- Clear all ARGB LED color

- Initialize SysTick as time counting

- Periodically calculate lighting mode and update ARGB LED data

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Sample Code

- System Init
  - Function SYS_Init
    - Initialize system clock

```
/* Set core clock to HCLK_CLK MHz */
CLK_SetCoreClock(HCLK_CLK);
 /* Update the Variable SystemCoreClock */
SystemCoreClockUpdate();
```

“Platform_HAL_NUC1262.h”

```
#define HCLK_CLK     72000000
```

➢ Enable PLL, set PLLFOUT frequency to HCLK_CLK*2, and set PLLFOUT/2 to HCLK.

    - Enable PDMA module

```
/* Enable module clock */
CLK_EnableModuleClock(PDMA_MODULE);
/* Reset PDMA module */
SYS_ResetModule(PDMA_RST);
```


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Sample Code

- System Init
  - Function SYS_Init
    - Check if Data Flash has LED setting.
    - If not, write the default setting.

```
/* Check Data Flash */
FMC_Open();
Temp = FMC_ReadDataFlashBaseAddr();
/* Write default setting if Data Flash is all 0xFFFFFFFF */
if(FMC_GetCheckSum(Temp, FMC_FLASH_PAGE_SIZE*NUM_DATAFLASH_PAGE) == CRC32_DATAFLASH_ALL_0XFF)
{
    WriteDefaultSetting();
}
```

Enable FMC module to operate Flash.

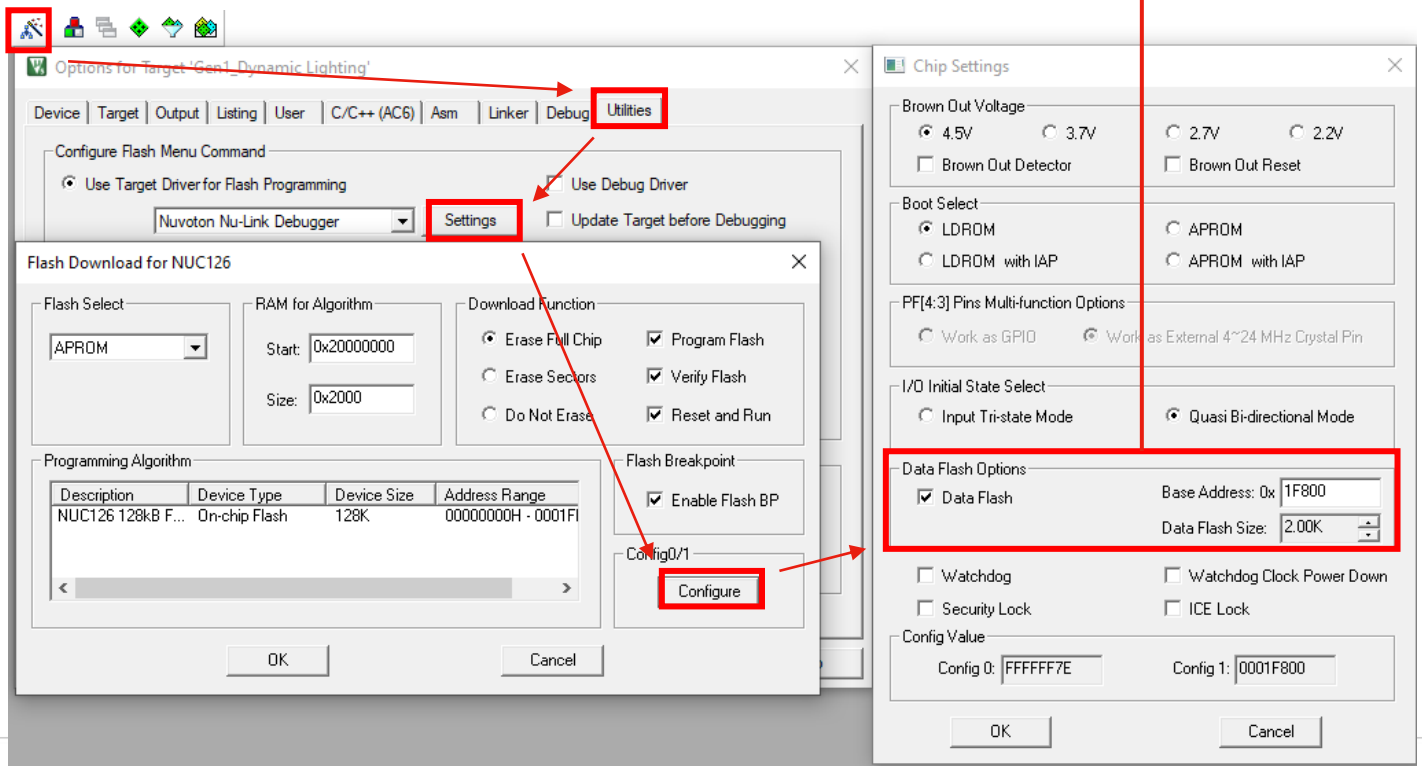
Read Data Flash base address.

➢ Use CRC32 to calculate checksum.
If the checksum value is the same as the one for all 0xFF, write the default LED setting.

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Sample Code

- System Init
  - Set User Configure to enable Data Flash.
    - Enable “Data Flash” and set the Data Flash Size.
      Target Options -> Utilities -> Settings -> Configure



“Flash.h”

```
/*------Define Data Flash for LED setting------*/
#define NUM_DATAFLASH_PAGE          1
```


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Sample Code

- System Init
  - Function ReadStoredSetting
    - Read LED setting from Data Flash.

```
for(i = 0; i < TOTAL_LED_AREA; i++)
{
    *(uint32_t *)&Temp1 = FMC_Read(Data_Address + LEN_LED_SETTING*i     );
    LED_Mapping[i]->AP_Sync      = Temp1.AP_Sync;
    LED_Mapping[i]->LEDNum       = Temp1.LED_Nums;
    LED_Mapping[i]->LightingMode = Temp1.LightingMode;
    LED_Mapping[i]->Mode_FUNC    = (LED_FUNC)Mode_Function[LED_Mapping[i]->LightingMode];
    *(uint32_t *)&Temp2 = FMC_Read(Data_Address + LEN_LED_SETTING*i +  4);
    LED_Mapping[i]->Color_R      = Temp2.Color_R;
    LED_Mapping[i]->Color_G      = Temp2.Color_G;
    LED_Mapping[i]->Color_B      = Temp2.Color_B;
    LED_Mapping[i]->Brightness   = Temp2.Brightness;
    *(uint32_t *)&Temp3 = FMC_Read(Data_Address + LEN_LED_SETTING*i +  8);
    LED_Mapping[i]->Speed        = Temp3.Speed;
    LED_Mapping[i]->Direction    = Temp3.Direction;
    LED_Mapping[i]->LED_Type     = Temp3.LED_Type;
}
```


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Sample Code

- Structure **LED_Setting_T**
  - Variables for lighting mode operation.
    - **"LED_Control.h"**

```
typedef struct LED_Setting_Tag
{
    uint32_t  TimeCounter;
    uint8_t   AP_Sync;
    uint8_t   LEDNum;
    uint8_t   LightingMode;
    uint8_t   Color_R;
    uint8_t   Color_G;
    uint8_t   Color_B;
    uint8_t   Brightness;
    uint8_t   Speed;
    uint8_t   Direction;
    uint8_t   LED_Type;
    uint8_t   fPDMA_Done;
    uint8_t   RandomFlag;
    uint32_t  Random;
    LED_FUNC  Mode_FUNC;
    uint8_t*  LED_Data;
    uint8_t   Main_Volume;
    uint8_t   Left_Volume;
    uint8_t   Right_Volume;
    uint8_t   Music_Action;
    uint32_t  Array_Size;
    uint16_t  LED_Offset;
    uint8_t   ColorIndex;
    uint8_t   Sequence;
    /* Dynamic Lighting */
    uint8_t   Dynamic_Update_Flag;
    uint16_t  Target_LampID;
    uint8_t   Autonomous_Flag;
    Lamp_RGBI_T*  DynamicData;
} LED_Setting_T;
```


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Sample Code

- Structure **LED_Setting_T**
  - Variables for lighting mode operation.
    - "LED_Control.h"

```
typedef struct LED_Setting_Tag
{
    uint32_t  TimeCounter;
    uint8_t   AP_Sync;
    uint8_t   LEDNum;
    uint8_t   LightingMode;
    uint8_t   Color_R;
    uint8_t   Color_G;
    uint8_t   Color_B;
    uint8_t   Brightness;
    uint8_t   Speed;
    uint8_t   Direction;
    uint8_t   LED_Type;
    uint8_t   fPDMA_Done;
    uint8_t   RandomFlag;
    uint32_t  Random;
    LED_FUNC  Mode_FUNC;
```

Store to Data Flash

Parameters' Description

TimeCounter	Base timer counter for lighting mode calculation to show effect. Use SysTick to count.
AP_Sync	LED color is controlled by PC tool.
LEDNum	Numbers of LED count.
LightingMode	Mode index for Mode_Function.
Color_R/G/B	RGB color setting for lighting mode. Max value is 255.
Brightness	Brightness setting for lighting mode. Max value is 255.
Speed	Speed setting for lighting mode. Max value is 255.
Direction	Direction setting for lighting mode, forward or backward.
LED_Type	LED data format type, RGB or GRB.
fPDMA_Done	Clear to 0 before trigger LLSI transmission. Set 1 in LLSI interrupt handler to indicate transmission done.
RandomFlag	For Random mode calculation.
Random	For Random mode calculation.
Mode_FUNC	Function pointer to call corresponding function based on index Lighting Mode.

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Sample Code

- Structure **LED_Setting_T**
  - Variables for lighting mode operation.
    - “LED_Control.h”

Parameters’ Description

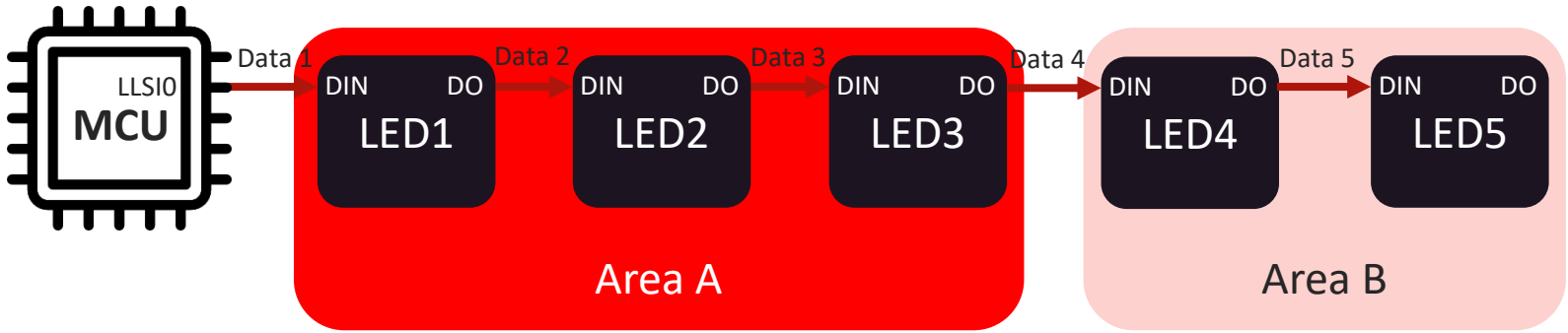
LED_Data	Pointer to LED RGB data.
Array_Size	Size of LED RGB data array.
LED_Offset	LED data offset for LLSI output data. Used for single LLSI to control multiple Gen1 ARGB LED area.
ColorIndex	Color pointer for Cycling, Random and Water mode.
Sequence	Index of LED_Module_Setting to trigger PDMA.

```
        uint8_t*  LED_Data;
        uint8_t   Main_Volume;
        uint8_t   Left_Volume;
        uint8_t   Right_Volume;
        uint8_t   Music_Action;
        uint32_t  Array_Size;
        uint16_t  LED_Offset;
        uint8_t   ColorIndex;
        uint8_t   Sequence;
        /* Dynamic Lighting */
        uint8_t   Dynamic_Update_Flag;
        uint16_t  Target_LampID;
        uint8_t   Autonomous_Flag;
        Lamp_RGBI_T*  DynamicData;
} LED_Setting_T;
```


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Sample Code

- LED_Offset
  - A.LED_Data = LLSI0_DATA;
    A.LED_Offset = 0;
  - B.LED_Data = LLSI0_DATA;
    B.LED_Offset = A.LEDNum * 3;



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Sample Code

- ARGB Port Init
  - Function ARGB_Port_Initial (HAL_ARGB_Port_Initial)
    - Initialize LLSI module.
      (Sample code uses mapping table, here take **Strip A** for example)

```
"Platform_HAL_NUC1262.h"
/*------ARGB LED Control Definition----------*/
/* Strip A */
#define LED_IF_NUM_STRIP_A    4
#define LED_IF_MODULE_STRIP_A LLSI4
#define LED_IF_IRQ_STRIP_A    LLSI4_IRQHandler
#define PDMA_CH_STRIP_A       0
/* Strip A */
#define IO_PORT_STRIP_A     PA
#define IO_NUM_STRIP_A      3
#define IO_BIT_STRIP_A      BIT3
#define IO_IRQ_STRIP_A      GPAB_IRQn
#define MFP_OFFSET_STRIP_A  0x30    // SYS->GPA_MFPL
#define PDIO_OFFSET_STRIP_A 0x0C    // PA.3
#define IO_GPIO_MSK_STRIP_A SYS_GPA_MFPL_PA3MFP_Msk
#define IO_MFP_MSK_STRIP_A  SYS_GPA_MFPL_PA3MFP_LLSI4_OUT
```

```
/* Enable module clock */
CLK_EnableModuleClock(LLSI4_MODULE);
/* Set LLSI multi-function pin */
SYS->GPA_MFPL = (SYS->GPA_MFPL & ~SYS_GPA_MFPL_PA3MFP_Msk) | SYS_GPA_MFPL_PA3MFP_LLSI4_OUT;
/* Set LLSI configuration */
LLSI_Open(LLSI4, LLSI_MODE_SW, LLSI_FORMAT_GRB, HCLK_CLK, ARGB_LED_PERIOD_NANOSECOND,
ARGB_LED_T0H_NANOSECOND, ARGB_LED_T1H_NANOSECOND, ARGB_LED_RESET_NANOSECOND, NUM_LED_STRIP,
LLSI_IDLE_LOW);
/* Enable Reset command function */
LLSI_ENABLE_RESET_COMMAND(LLSI4);
/* Enable Reset command interrupt */
LLSI_EnableInt(LLSI4, LLSI_RST_INT_MASK);
/* Enable NVIC for LLSI */
NVIC_EnableIRQ(LLSI4_IRQn)
```


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Sample Code

- ARGB Port Init
  - Function PDMA_Initial (HAL_PDMA_Initial)
    - Initialize PDMA channel.
      (Sample code uses mapping table, here take **Strip A** for example)

```
/* Open Channel */
PDMA_Open(1 << 0);
/* PDMA Setting for LLSI */
/* Transfer type is single transfer */
PDMA_SetBurstType(0, PDMA_REQ_SINGLE, 0);
/* Set source address, destination address */
PDMA_SetTransferAddr(0, LEDSetting_Strip_A.LED_Data, PDMA_SAR_INC, (uint32_t)&(LLSI4->DATA),
PDMA_DAR_FIX);
```

Source Address.

Destination Address.

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Sample Code

- ARGB Port Init
  - Function Clear_LED_Data
    - Clear all ARGB LED color.

```
/* Set all LED data to 0x0 */
for(i = 0; i < &LEDSetting_Strip_A->Array_Size; i++)
{
    &LEDSetting_Strip_A->LED_Data[i] = 0x0;
}
/* Trigger LLSI and PDMA to transfer data out */
Set_LED_Data(&LEDSetting_Strip_A);
```

  - Function Set_LED_Data (HAL_Set_LED_Data)
    - Trigger LLSI and PDMA to transfer data out.

```
/* Clear done flag */
&LEDSetting_Strip_A->fPDMA_Done = 0;
/* Calculate transfer count */
u32Count = &LEDSetting_Strip_A->Array_Size / 4; if(&LEDSetting_Strip_A->Array_Size % 4) u32Count++;
/* Set transfer count */
PDMA_SetTransferCnt(0, PDMA_WIDTH_32, u32Count);
/* Set request source */
PDMA_SetTransferMode(0, PDMA_LLSI4, FALSE, 0);
/* Change LLSI mode */
LLSI_SET_PDMA_MODE(LLS4);
```

Transfer count should be **word alignment**.
For example, 10 pcs LED need 30 bytes data.
Then transfer count should be 8.

LLSI data width is 32 bits.

PDMA trigger source.

Sample Code

- ARGB Port Init
  - Function LLSIx_IRQHandler
    - After LLSI transfers Reset Code, LLSI interrupt causes IRQHandler.

```
if(!LEDSetting_Strip_A.fPDMA_Done)
{
    LEDSetting_Strip_A.fPDMA_Done = 1;
    LLSI_SET_SW_MODE(LLSI4);
}
/* Clear interrupt flag */
LLSI_ClearIntFlag(LLSI4, LLSI_RST_INT_MASK | LLSI_FEND_INT_MASK);
```

Set LLSI to software mode to stop sending request to PDMA.

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Sample Code

- SysTick Init
  - Function SysTick_Initial
    - Use SysTick to add TimeCounter every millisecond.

```
/* Set SysTick clock source to HCLK */
SysTick->CTRL = SysTick_CTRL_CLKSOURCE_Msk;
/* Set load value to make SysTick period is 1 ms */
SysTick->LOAD = HCLK_CLK / 1000;
/* Clear SysTick counter */
SysTick->VAL = 0;
/* Enable SysTick interrupt */
SysTick->CTRL |= SysTick_CTRL_TICKINT_Msk;
/* Enable SysTick NVIC */
NVIC_EnableIRQ(SysTick_IRQn);
/* Start SysTick */
SysTick->CTRL |= SysTick_CTRL_ENABLE_Msk;
```


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Sample Code

- SysTick Init
  - Function SysTick_Handler
    - SysTick triggers interrupt handler every millisecond.

```
/* Add LED mode counter */
TimeCounterFrameUpdate++;
for(i = 0; i < TOTAL_LED_AREA; i++)
    LED_Mapping[i]->TimeCounter++;
/* Update LED frame data */
if(TimeCounterFrameUpdate >= TIMER_FRAME_UPDATE)
{
    LED_Frame_Update_flag = 1;
    TimeCounterFrameUpdate -= TIMER_FRAME_UPDATE;
}
```

“**LED_Control.h**”

```
#define TIMER_FRAME_UPDATE 25
```

➢ Time to change LED color.
Unit in millisecond.

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Sample Code

- Main Loop
  - Periodically calculate lighting mode and update ARGB LED data.

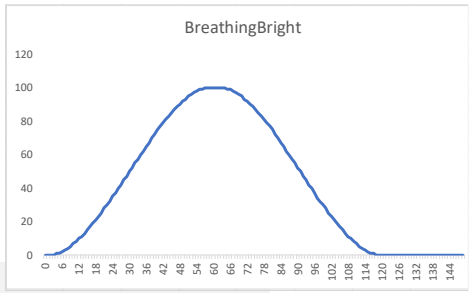
```
/* Update LED Control */
if(LED_Frame_Update_flag)
{
    for(target_port = 0; target_port < TOTAL_LED_AREA; target_port++)
    {
        /* Check if data transmission is done */
        if(LED_Mapping[target_port]->fPDMA_Done == 1)
        {
            if(LED_Mapping[target_port]->AP_Sync == FALSE)
            {
                for(i = 0; i < LED_Mapping[target_port]->Array_Size; i++)
                {
                    *(LED_Mapping[target_port]->LED_Data + i) = 0x0;
                }
                /* LED data */
                LED_Mapping[target_port]->Mode_FUNC(LED_Mapping[target_port]);
            }
            /* Set Serial Data */
            Set_LED_Data(LED_Mapping[target_port]);
        }
    }
    /* Clear flag */
    LED_Frame_Update_flag = 0;
}
```

Call function to calculate the LED data based on TimeCounter value for target lighting mode.

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Sample Code

- Main Loop
  - Function FUNC_Breathing
    - Calculate LED data in Breathing mode.



“LED_Control.c”

```
#define BreathingArraySize 150
const uint8_t BreathingBright[BreathingArraySize];
```

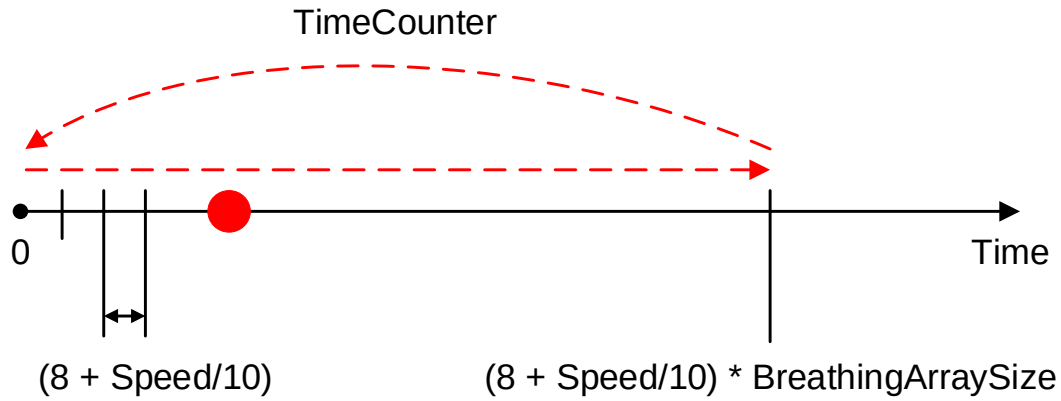
Lookup table for Breathing mode.

```
/* Calculate Color */
Temp = HDIV_Mod(HDIV_Div(LED_Setting->TimeCounter, 8 + HDIV_Div(LED_Setting->Speed, 10)), BreathingArraySize);
TempR = HDIV_Div((LED_Setting->Color_R * LED_Setting->Brightness * *(BreathingBright + Temp)), 0xFF*100);
TempG = HDIV_Div((LED_Setting->Color_G * LED_Setting->Brightness * *(BreathingBright + Temp)), 0xFF*100);
TempB = HDIV_Div((LED_Setting->Color_B * LED_Setting->Brightness * *(BreathingBright + Temp)), 0xFF*100);
/* Mapping Color to LED Format */
Set_Single(LED_Setting->LED_Data, LED_Setting->LEDNum, LED_Setting->LED_Offset, TempR, TempG, TempB);
/* Reset CountingTime */
if(LED_Setting->TimeCounter >= ((8 + HDIV_Div(LED_Setting->Speed, 10)) * BreathingArraySize))
{
    LED_Setting->TimeCounter -= (8 + HDIV_Div(LED_Setting->Speed, 10)) * BreathingArraySize;
}
```


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Sample Code

- Main Loop
  - Function FUNC_Breathing
    - Calculate LED data in Breathing mode.




```
/* Calculate Color */
Temp = HDIV_Mod(HDIV_Div(LED_Setting->TimeCounter, 8 + HDIV_Div(LED_Setting->Speed, 10)), BreathingArraySize);
TempR = HDIV_Div((LED_Setting->Color_R * LED_Setting->Brightness * *(BreathingBright + Temp)), 0xFF*100);
TempG = HDIV_Div((LED_Setting->Color_G * LED_Setting->Brightness * *(BreathingBright + Temp)), 0xFF*100);
TempB = HDIV_Div((LED_Setting->Color_B * LED_Setting->Brightness * *(BreathingBright + Temp)), 0xFF*100);
/* Mapping Color to LED Format */
Set_Single(LED_Setting->LED_Data, LED_Setting->LEDNum, LED_Setting->LED_Offset, TempR, TempG, TempB);
/* Reset CountingTime */
if(LED_Setting->TimeCounter >= ((8 + HDIV_Div(LED_Setting->Speed, 10)) * BreathingArraySize))
{
    LED_Setting->TimeCounter -= (8 + HDIV_Div(LED_Setting->Speed, 10)) * BreathingArraySize;
}
```

Execution time for single effect cycle.

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Sample Code

- Main Loop
  - Function FUNC_Breathing
    - Calculate LED data in Breathing mode.

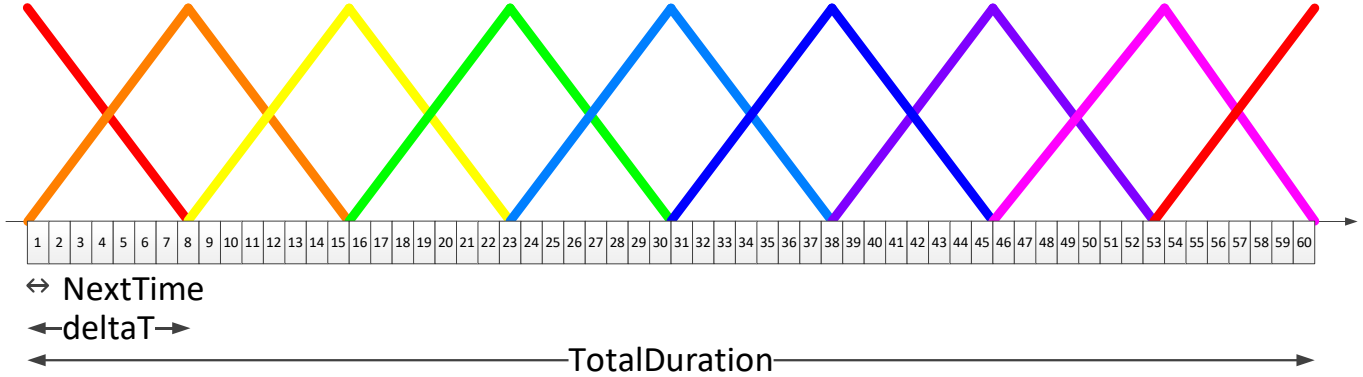
```
/* Calculate Color */
Temp = HDIV_Mod(HDIV_Div(LED_Setting->TimeCounter, 8 + HDIV_Div(LED_Setting->Speed, 10)), BreathingArraySize);
TempR = HDIV_Div((LED_Setting->Color_R * LED_Setting->Brightness * *(BreathingBright + Temp)), 0xFF*100);
TempG = HDIV_Div((LED_Setting->Color_G * LED_Setting->Brightness * *(BreathingBright + Temp)), 0xFF*100);
TempB = HDIV_Div((LED_Setting->Color_B * LED_Setting->Brightness * *(BreathingBright + Temp)), 0xFF*100);
/* Mapping Color to LED Format */
Set_Single(LED_Setting->LED_Data, LED_Setting->LEDNum, LED_Setting->LED_Offset, TempR, TempG, TempB);
/* Reset CountingTime */
if(LED_Setting->TimeCounter >= ((8 + HDIV_Div(LED_Setting->Speed, 10)) * BreathingArraySize))
{
    LED_Setting->TimeCounter -= (8 + HDIV_Div(LED_Setting->Speed, 10)) * BreathingArraySize;
}
```

Set calculated LED RGB data to array for LLSI to transfer.

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Sample Code

- Main Loop
  - Function FUNC_Rainbow
    - Calculate timing in Rainbow mode.




```
/* Duration for one time effect */
TotalDuration = HDIV_Div(60 * 1000, HDIV_Div(255 - LED_Setting->Speed, 3) + 5);
/* Next LED time */
NextTime = HDIV_Div(TotalDuration, 60);
/* Current Time */
CurrentTime = HDIV_Mod(LED_Setting->TimeCounter, TotalDuration);
/* Unit time for each color */
deltaT = HDIV_Div(TotalDuration, RainbowSize);

/* Reset CountingTime */
if(LED_Setting->TimeCounter >= TotalDuration)
{
    LED_Setting->TimeCounter -= TotalDuration;
}
```


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Sample Code

- Main Loop
  - Function FUNC_Rainbow
    - Calculate timing in Rainbow mode.

“LED_Control.c”

```
#define RainbowSize 8
const uint8_t RainbowColor[RainbowSize][3]    // (R, G, B)
              = {{255,   0,   0},   // Red
                 {255,  85,   0},   // Orange
                 {255, 255,   0},   // Yellow
                 {  0, 255,   0},   // Green
                 {  0, 127, 255},   // Cyan
                 {  0,   0, 255},   // Blue
                 {127,   0, 255},   // Indigo
                 {255,   0, 255}};  // Purple
```

Color table for Rainbow.

```
/* Duration for one time effect */
TotalDuration = HDIV_Div(60 * 1000, HDIV_Div(255 - LED_Setting->Speed, 3) + 5);
/* Next LED time */
NextTime = HDIV_Div(TotalDuration, 60);
/* Current Time */
CurrentTime = HDIV_Mod(LED_Setting->TimeCounter, TotalDuration);
/* Unit time for each color */
deltaT = HDIV_Div(TotalDuration, RainbowSize);

/* Reset CountingTime */
if(LED_Setting->TimeCounter >= TotalDuration)
{
    LED_Setting->TimeCounter -= TotalDuration;
}
```


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Sample Code

- Main Loop
  - Function FUNC_Rainbow
    - Calculate LED data in Rainbow mode.

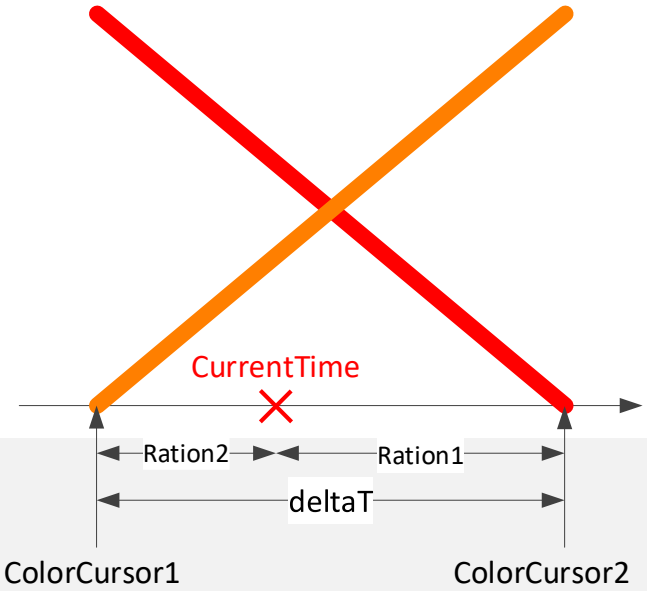
To avoid the situation that *TotalDuration > CurrentTime > (deltaT * RainbowSize)* due to quotient calculation.

```
/* Mix Color */
for(j = 0; j < LED_Setting->LEDNum; j++, CurrentTime += NextTime) {
    if(CurrentTime >= TotalDuration)
        CurrentTime -= TotalDuration;
    if(CurrentTime >= (deltaT * RainbowSize)) {
        /* Color Cursor */
        ColorCursor1 = 0;
        ColorCursor2 = 1;
        for(i = 0; i < 3; i++) {
            DisplayColor[LED_Setting->LEDNum-j-1][i] = RainbowColor[ColorCursor1][i];
        }
    } else {
        /* Color Cursor */
        ColorCursor1 = HDIV_Mod(HDIV_Div(CurrentTime, deltaT), RainbowSize);
        ColorCursor2 = HDIV_Mod((ColorCursor1 + 1), RainbowSize);
        for(i = 0; i < 3; i++) {
            DisplayColor[LED_Setting->LEDNum-j-1][i] = HDIV_Div(((RainbowColor[ColorCursor1][i] * ((ColorCursor1 + 1) * deltaT - CurrentTime)) + \
                                                                 (RainbowColor[ColorCursor2][i] * (CurrentTime - ColorCursor1 * deltaT))), deltaT);
        }
    }
}
```


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Sample Code

- Main Loop
  - Function FUNC_Rainbow
    - Calculate LED data in Rainbow mode.



$$Color = \frac{Color1 \times Ration1}{deltaT} + \frac{Color2 \times Ration2}{deltaT}$$

$$= \frac{Color1 \times \left((ColorCursor1 + 1) \times deltT - CurrentTime\right) + Color2 \times (CurrentTime - ColorCurcor1 \times deltaT)}{deltaT}$$

```
/* Mix Color */
for(j = 0; j < LED_Setting->LEDNum; j++, CurrentTime += NextTime) {
    if(CurrentTime >= TotalDuration)
        CurrentTime -= TotalDuration;
    if(CurrentTime >= (deltaT * RainbowSize)) {
        /* Color Cursor */
        ColorCursor1 = 0;
        ColorCursor2 = 1;
        for(i = 0; i < 3; i++) {
            DisplayColor[LED_Setting->LEDNum-j-1][i] = RainbowColor[ColorCursor1][i];
        }
    } else {
        /* Color Cursor */
        ColorCursor1 = HDIV_Mod(HDIV_Div(CurrentTime, deltaT), RainbowSize);
        ColorCursor2 = HDIV_Mod((ColorCursor1 + 1), RainbowSize);
        for(i = 0; i < 3; i++) {
            DisplayColor[LED_Setting->LEDNum-j-1][i] = HDIV_Div(((RainbowColor[ColorCursor1][i] * ((ColorCursor1 + 1) * deltaT - CurrentTime)) + \
                                                                 (RainbowColor[ColorCursor2][i] * (CurrentTime - ColorCursor1 * deltaT))), deltaT);
        }
    }
}
```


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Sample Code

- Main Loop
  - Function FUNC_Rainbow
    - Calculate LED data in Rainbow mode.

```
/* Mapping Color to LED Format */
/* Direction forward or backward */
if(LED_Setting->Direction == eDir_Forward)
    Set_Array(LED_Setting->LED_Data, LED_Setting->LEDNum, LED_Setting->LED_Offset, (uint8_t *)DisplayColor,
LED_Setting->Brightness);
else if(LED_Setting->Direction == eDir_Backward)
    Set_InverseArray(LED_Setting->LED_Data, LED_Setting->LEDNum, LED_Setting->LED_Offset, (uint8_t
*)DisplayColor, LED_Setting->Brightness);
```

Set calculated DisplayColor data to array for LLSI to transfer.

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Sample Code

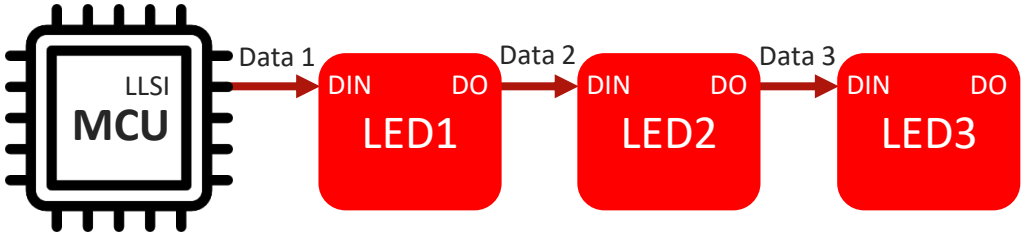
- Main Loop
  - Function Set_Single
    - Set **single** RGB color data to **all LED**.
    - Need to **calculate brightness** first.

```
for(i = 0; i < TotalLED; i++)
{
    *(LED_DATA + Offset + i*3 + 0) = Data_R;
    *(LED_DATA + Offset + i*3 + 1) = Data_G;
    *(LED_DATA + Offset + i*3 + 2) = Data_B;
}
```

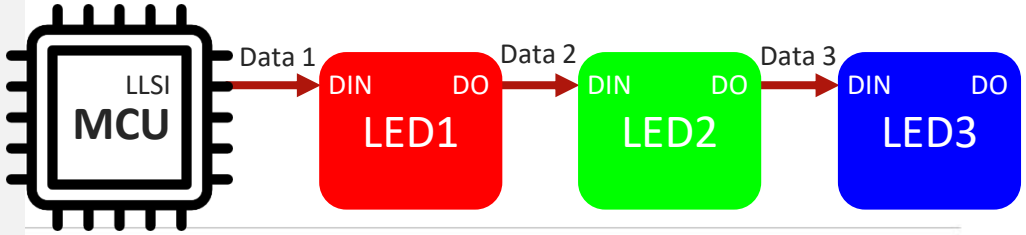
  - Function Set_Array
    - Set **multiple** RGB color to **each** LED.

```
for(i = 0; i < TotalLED; i++)
{
    TempR = HDIV_Div(*(DisplayData + i*3 + 0) * MaxBright, 0xFF);
    TempG = HDIV_Div(*(DisplayData + i*3 + 1) * MaxBright, 0xFF);
    TempB = HDIV_Div(*(DisplayData + i*3 + 2) * MaxBright, 0xFF);
    *(LED_DATA + Offset + i*3 + 0) = TempR;
    *(LED_DATA + Offset + i*3 + 1) = TempG;
    *(LED_DATA + Offset + i*3 + 2) = TempB;
}
```

- {R, G, B} = {255, 0, 0}



- {R1, G1, B1} = {255, 0, 0}
  {R2, G2, B2} = {0, 255, 0}
  {R3, G3, B3} = {0, 0, 255}



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Sample Code

- Main Loop
  - Periodically calculate lighting mode and update ARGB LED data.

```
/* Update LED Control */
if(LED_Frame_Update_flag)
{
    for(target_port = 0; target_port < TOTAL_LED_AREA; target_port++)
    {
        /* Check if data transmission is done */
        if(LED_Mapping[target_port]->fPDMA_Done == 1)
        {
            if(LED_Mapping[target_port]->AP_Sync == FALSE)
            {
                for(i = 0; i < LED_Mapping[target_port]->Array_Size; i++)
                {
                    *(LED_Mapping[target_port]->LED_Data + i) = 0x0;
                }
                /* LED data */
                LED_Mapping[target_port]->Mode_FUNC(LED_Mapping[target_port]);
            }
            /* Set Serial Data */
            Set_LED_Data(LED_Mapping[target_port]);
        }
    }
    /* Clear flag */
    LED_Frame_Update_flag = 0;
```


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DEMO



Joy of innovation

NUVOTON

Thank You

Danke

Merci

ありがとう

Gracias

Kiitos

감사합니다

धन्यबाद

شكرا لك

הדות